Mesoscopic additive spikes for rectangular random matrices

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Abstract

Let

$$W_n = X_n + P_n,$$

where $X_n \in \mathbb{F}^{n\times m}$ is rectangular random noise with $n/m \to c \in (0,1]$ and P_n has rank $r_n \to \infty$ with $r_n = o(n)$. We study the singular values of W_n in the *mesoscopic-rank* regime.

The paper is organized around an abstract input: a uniform operator-norm bound for the $r_n \times r_n$ resolvent compressions associated with the spike singular vectors. Assume, in addition, that the squared singular-value empirical law of X_n converges to a compactly supported probability measure μ with right edge b^2, that $\|X_n\|_{\mathrm{op}} \to b$ in probability, and that the associated rectangular D-transform satisfies the finite positive edge condition $d(b^+) \in (0,\infty)$. Then:

1. For every fixed $\delta > 0$, all spikes above the BBP threshold by at least δ are matched *in order* with the corresponding outlier locations $\rho(\theta)$, uniformly over all such indices, without any separation assumption among those spikes.
2. On the same scale there are no additional singular values above the supercritical window: if

$$k_{n,\delta} := \#\{i : \theta_{n,i} \ge \theta_\star + \delta\},$$

then

$$\sigma_{k_{n,\delta}+1}(W_n) \le \rho(\theta_\star + \delta) + o_{\mathbb{P}}(1).$$

3. For every fixed $\varepsilon > 0$, if the spike empirical measures satisfy

$$H_n := \frac{1}{r_n}\sum_{i=1}^{r_n} \delta_{\theta_{n,i}} \Rightarrow H \qquad \text{weakly in probability,}$$

then the empirical measure of the *actual* eigenvalues of $W_n^* W_n$ above $(b+\varepsilon)^2$, normalized by r_n, converges vaguely in probability on $((b+\varepsilon)^2, \infty)$ to the restricted pushforward $(\Lambda_\# H)|_{((b+\varepsilon)^2,\infty)}$. Equivalently, the singular-value point measure above $b+\varepsilon$ converges vaguely to $(\rho_\# H)|_{(b+\varepsilon,\infty)}$. Under the natural no-boundary-mass condition at $(b+\varepsilon)^2$, this upgrades to weak convergence of finite measures and yields convergence of the normalized outlier count.

Once the scalar convergence and compression hypothesis are available, the outlier analysis is deterministic.

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Contents

1 Introduction

1.1 Model

Let $X_n \in \mathbb{F}^{n\times m}$ with $n \le m$ be random, and let $P_n \in \mathbb{F}^{n\times m}$ be deterministic or independent of X_n. We study

$$W_n := X_n + P_n.$$

Write the singular values of W_n in decreasing order as

$$\sigma_1(W_n) \ge \cdots \ge \sigma_n(W_n) \ge 0,$$

and the eigenvalues of $W_n^* W_n$ in decreasing order as

$$\lambda_1(W_n^* W_n) \ge \cdots \ge \lambda_m(W_n^* W_n) \ge 0.$$

Of course $\lambda_i(W_n^* W_n) = \sigma_i(W_n)^2$ for $1 \le i \le n$, and the remaining $m - n$ eigenvalues are zero.

We are interested in the *mesoscopic-rank* regime

$$r_n := \operatorname{rank}(P_n) \to \infty, \qquad r_n = o(n).$$

This regime lies between fixed rank, where one sees finitely many outliers, and macroscopic rank, where the perturbation changes the global limiting spectral distribution.

1.2 What is proved here

The main point of the paper is to separate the deterministic outlier mechanism from the model-dependent probabilistic input. The abstract theorem requires only two ingredients: a deterministic limiting law for the noise singular values, and a uniform control of the resolvent compressions along the spike directions. Once these are available, the remaining outlier analysis is deterministic.

More concretely:

- Sections 2–5 prove the main theorems under an abstract *resolvent-compression hypothesis*.
- The ordered matching theorem is uniform over all spikes separated from the threshold by a fixed $\delta > 0$, and it does *not* require any mutual separation among those spikes.
- Section 7 verifies the compression hypothesis for Haar-singular-vector ensembles, at the level of the joint law, and includes rectangular Gaussian noise.
- Section 8 records the explicit Marchenko–Pastur formulas [17].
- For Haar-singular-vector ensembles, the net argument in the compression proof yields the quantitative rate

$$\beta_n(I) \lesssim \sqrt{\frac{r_n}{n}} + \sqrt{\frac{\log n}{n}}.$$

The resulting theory yields, in the regime $r_n = o(n)$, a uniform ordered matching theorem for all spikes at a fixed distance above threshold, a matching no-extra-outliers statement on the same scale, and a spike-scale law for the actual outlier spectrum above any fixed separation from the edge.

1.3 Position relative to the literature

Fixed-rank rectangular additive deformations are classical: the outlier locations are described by the rectangular D-transform in Benaych-Georges–Nadakuditi [5], while the BBP terminology originates in the spiked covariance transition of Baik–Ben Arous–Péché [3]. Gaussian information-plus-noise localization and fixed-rank outlier analysis were developed by Loubaton–Vallet and by Chapon–Couillet–Hachem–Mestre [16, 7], while exact separation for spiked information-plus-noise models was developed by Capitaine [8]. For covariance-type models and principal components under local-law inputs, see for example Bloemendal–Erdős–Knowles–Yau–Yin, Bloemendal–Knowles–Yau–Yin, and Knowles–Yin [4, 6, 13]; for deformed rectangular models with rates and denoising applications, see Ding [9]. Huang treated mesoscopic perturbations in Hermitian additive and multiplicative settings [12]. More recent growing-rank rectangular signal-plus-noise results in different models and regimes include Liu–Liu–Pan–Zhang–Zhang [15], while large-rank edge statistics were analyzed by Ding–Yang [10]. On the physics and inference side, high-rank additive rectangular models have also been studied in the "extensive spike" regime by Landau–Mel–Ganguli [14].

The present paper occupies a narrower and more specific niche: it treats *rectangular additive* deformations with *sublinear rank* $r_n = o(n)$ under an abstract resolvent-compression hypothesis, proves an ordered matching theorem uniform over all spikes at distance δ above threshold without any spike-separation assumption, and verifies the hypothesis completely for Haar-singular-vector ensembles. The paper should therefore be viewed as a rectangular mesoscopic complement to Huang's Hermitian theory, rather than as a substitute for the broader fixed-rank, local-law, denoising, or large-rank literatures.

2 Setup and assumptions

2.1 Signal model

Assumption 2.1 (Signal)**.** Let $P_n \in \mathbb{F}^{n\times m}$ have rank r_n with

$$r_n \to \infty, \qquad r_n = o(n).$$

Fix a measurable choice of thin singular-value decomposition

$$P_n = U_n \Theta_n V_n^*,$$

where $U_n \in \mathbb{F}^{n\times r_n}$ and $V_n \in \mathbb{F}^{m\times r_n}$ have orthonormal columns, and

$$\Theta_n = \operatorname{diag}(\theta_{n,1}, \dots, \theta_{n,r_n}), \qquad \theta_{n,1} \ge \cdots \ge \theta_{n,r_n} > 0.$$

Assume

$$\sup_{n,i} \theta_{n,i} \le \theta_{\max} < \infty.$$

If P_n is random, we assume it is independent of X_n.

Define the spike empirical measure

$$H_n := \frac{1}{r_n} \sum_{i=1}^{r_n} \delta_{\theta_{n,i}}.$$

2.2 Noise assumptions

Assumption 2.2 (Noise ensemble)**.** Let $X_n \in \mathbb{F}^{n\times m}$ with $n/m \to c \in (0,1]$. Assume that X_n is independent of P_n and that the following hold.

1. There exists a compactly supported probability measure μ on $[0,\infty)$ with right edge
$$b^2 := \sup \operatorname{supp}(\mu)$$
such that the empirical spectral distribution of $X_n X_n^*$ converges weakly to μ in probability and
$$\|X_n\|_{\mathrm{op}} \to b \qquad \text{in probability.}$$

2. Define, for $z > b$,
$$\varphi(z) := \int \frac{z}{z^2 - x}\,\mu(dx), \qquad \widetilde{\varphi}(z) := c\,\varphi(z) + (1-c)\frac{1}{z}, \qquad d(z) := \varphi(z)\widetilde{\varphi}(z).$$
Assume that
$$d(b^+) := \lim_{z\downarrow b} d(z) \in (0,\infty).$$

Remark 2.3. Assumption 2.2 is intentionally mild on the global side, but the edge condition $d(b^+) \in (0,\infty)$ is a genuine extra hypothesis. It holds in the Marchenko–Pastur case and, more generally, at standard regular right edges with no atom at b^2. The model-dependent difficulty is *not* in these scalar quantities, but in the uniform compression estimate introduced below.

For $z > \|X_n\|_{\mathrm{op}}$, define

$$\begin{aligned}
\varphi_n(z) &:= \frac{1}{n}\operatorname{tr}\Big(z(z^2\mathrm{I}_n - X_nX_n^*)^{-1}\Big),\\
\widetilde{\varphi}_n(z) &:= \frac{1}{m}\operatorname{tr}\Big(z(z^2\mathrm{I}_m - X_n^*X_n)^{-1}\Big) = \frac{n}{m}\varphi_n(z) + \frac{m-n}{m}\frac{1}{z},\\
d_n(z) &:= \varphi_n(z)\widetilde{\varphi}_n(z).
\end{aligned}$$

Remark 2.4 (Domain convention). If $I \subset\subset (b,\infty)$, then Assumption 2.2 implies
$$\mathbb{P}\big(\|X_n\|_{\mathrm{op}} < \inf I\big) \to 1.$$
Accordingly, whenever a random quantity initially defined only for $z > \|X_n\|_{\mathrm{op}}$ appears under $\sup_{z\in I}$, we tacitly work on the event $\{\|X_n\|_{\mathrm{op}} < \inf I\}$. Intersecting any high-probability event with this domain event changes its probability by at most $o(1)$.

Lemma 2.5 (Monotonicity and a derivative lower bound)**.** *For every n, the functions φ_n, $\widetilde{\varphi}_n$, and d_n are strictly decreasing on $(\|X_n\|_{\mathrm{op}},\infty)$. More precisely, if $I = [z_-, z_+]$ with $z_- > \|X_n\|_{\mathrm{op}}$, then for every $z \in I$,*
$$-\varphi_n'(z) \ge \frac{1}{z^2}, \qquad -\widetilde{\varphi}_n'(z) \ge \frac{1}{z^2}, \qquad -d_n'(z) \ge \frac{2}{z^3} \ge \frac{2}{z_+^3}.$$
The deterministic functions φ, $\widetilde{\varphi}$, and d are positive and strictly decreasing on (b,∞). Moreover, for every $z > b$,
$$-\varphi'(z) \ge \frac{1}{z^2}, \qquad -\widetilde{\varphi}'(z) \ge \frac{1}{z^2}, \qquad -d'(z) \ge \frac{2}{z^3}.$$
Finally, $d(z) \to 0$ as $z \to \infty$.

Proof. For $s \geq 0$ define

$$f_s(z) := \frac{z}{z^2 - s}, \qquad z > \sqrt{s}.$$

Then

$$f_s'(z) = -\frac{z^2 + s}{(z^2 - s)^2} < 0,$$

so each summand in φ_n and $\widetilde{\varphi}_n$ is strictly decreasing; hence φ_n, $\widetilde{\varphi}_n$, and $d_n = \varphi_n \widetilde{\varphi}_n$ are strictly decreasing on $(\|X_n\|_{\text{op}}, \infty)$.

For the derivative lower bounds, write

$$h_z(s) := \frac{z^2 + s}{(z^2 - s)^2}, \qquad s \in [0, z^2).$$

A direct differentiation gives

$$h_z'(s) = \frac{3z^2 + s}{(z^2 - s)^3} > 0,$$

so h_z is increasing in s and therefore

$$h_z(s) \geq h_z(0) = \frac{1}{z^2}.$$

Since

$$-\frac{d}{dz}\left(\frac{z}{z^2 - s}\right) = h_z(s),$$

averaging over the empirical singular values yields

$$-\varphi_n'(z) \geq \frac{1}{z^2}.$$

The same argument, together with the derivative of the extra $(m - n)/(mz)$ term, yields

$$-\widetilde{\varphi}_n'(z) \geq \frac{1}{z^2}.$$

Since also $\varphi_n(z) \geq 1/z$ and $\widetilde{\varphi}_n(z) \geq 1/z$, the product rule gives

$$-d_n'(z) = (-\varphi_n'(z))\widetilde{\varphi}_n(z) + \varphi_n(z)(-\widetilde{\varphi}_n'(z)) \geq \frac{2}{z^3}.$$

For the deterministic transforms, the same pointwise computation gives, for every $z > b$,

$$-\varphi'(z) = \int \frac{z^2 + x}{(z^2 - x)^2}\, \mu(dx) \geq \frac{1}{z^2}.$$

The identity $\widetilde{\varphi}(z) = c\varphi(z) + (1 - c)z^{-1}$ gives

$$-\widetilde{\varphi}'(z) = c(-\varphi'(z)) + (1 - c)z^{-2} \geq \frac{1}{z^2}.$$

Also $\varphi(z) \geq z^{-1}$ and $\widetilde{\varphi}(z) \geq z^{-1}$. Hence

$$-d'(z) = (-\varphi'(z))\widetilde{\varphi}(z) + \varphi(z)(-\widetilde{\varphi}'(z)) \geq \frac{2}{z^3}.$$

In particular, φ, $\widetilde{\varphi}$, and d are positive and strictly decreasing on (b, ∞). Finally, $d(z) \to 0$ because $\varphi(z) \sim 1/z$ and $\widetilde{\varphi}(z) \sim 1/z$ as $z \to \infty$. $\square$

Definition 2.6 (Threshold and outlier map)**.** Define the critical threshold

$$\theta_\star := d(b^+)^{-1/2}.$$

Since d is strictly decreasing from $d(b^+)$ to 0, for every $\theta > \theta_\star$ there is a unique $\rho(\theta) > b$ such that

$$d(\rho(\theta)) = \frac{1}{\theta^2}.$$

Set

$$\rho(\theta) := \begin{cases} d^{-1}(\theta^{-2}), & \theta > \theta_\star, \\ b, & \theta \le \theta_\star, \end{cases} \qquad \Lambda(\theta) := \rho(\theta)^2.$$

Lemma 2.7 (Basic properties of ρ)**.** *The map ρ is continuous on $(0,\infty)$ and strictly increasing on $(\theta_\star,\infty)$. Moreover, $\rho(\theta) > b$ if and only if $\theta > \theta_\star$.*

Proof. Strict monotonicity on $(\theta_\star,\infty)$ is immediate from the strict monotonicity of d on (b,∞). If $\theta \downarrow \theta_\star$, then $\theta^{-2} \uparrow d(b^+)$, and therefore

$$\rho(\theta) = d^{-1}(\theta^{-2}) \downarrow b.$$

Thus the right limit at $\theta_\star$ equals the value prescribed for $\theta \le \theta_\star$, so ρ is continuous at $\theta_\star$. Continuity away from $\theta_\star$ is immediate. The characterization $\rho(\theta) > b \iff \theta > \theta_\star$ follows from the definition. □

Proposition 2.8 (Uniform scalar convergence away from the edge)**.** *Under Assumption 2.2, for every compact interval $I \subset\subset (b,\infty)$,*

$$\sup_{z\in I} |\varphi_n(z) - \varphi(z)| \to 0, \qquad \sup_{z\in I} |\widetilde{\varphi}_n(z) - \widetilde{\varphi}(z)| \to 0, \qquad \sup_{z\in I} |d_n(z) - d(z)| \to 0$$

in probability.

Proof. Fix $I = [b+\varepsilon, L]$ with $\varepsilon > 0$. Let $\mu_n := F^{X_n X_n^*}$ be the empirical spectral distribution of $X_n X_n^*$. By Assumption 2.2, every subsequence has a further subsequence—which we continue to denote by n—such that

$$\mu_n \Rightarrow \mu \qquad \text{and} \qquad \|X_n\|_{\mathrm{op}} \to b$$

almost surely. It is therefore enough to prove the stated convergence almost surely along such a subsequence.

On the almost-sure event just described, for all large n we have

$$\|X_n\|_{\mathrm{op}} \le b + \frac{\varepsilon}{2},$$

hence $\mathrm{supp}(\mu_n) \subset [0, M]$ with $M := (b+\varepsilon/2)^2$. For $z \in I$ define

$$f_z(x) := \frac{z}{z^2 - x}, \qquad x \in [0, M].$$

The map $(z,x) \mapsto f_z(x)$ is continuous on the compact set $I \times [0,M]$, so the family $\{f_z : z \in I\}$ is uniformly bounded and uniformly continuous. Fix $\eta > 0$. By uniform continuity there exists a finite set

$$\mathcal{Z}_\eta \subset I$$

such that for every $z \in I$ there exists $z_0 \in \mathcal{Z}_\eta$ with

$$\sup_{x\in[0,M]} |f_z(x) - f_{z_0}(x)| \le \eta.$$

Therefore

$$\sup_{z\in I} \left| \int f_z \, d(\mu_n - \mu) \right| \le \max_{z_0\in\mathcal{Z}_\eta} \left| \int f_{z_0} \, d(\mu_n - \mu) \right| + 2\eta.$$

Since $\mu_n \Rightarrow \mu$ almost surely, the finite maximum on the right tends to 0 almost surely. Because $\eta > 0$ is arbitrary,

$$\sup_{z\in I} |\varphi_n(z) - \varphi(z)| \to 0 \qquad \text{almost surely.}$$

The identity

$$\widetilde{\varphi}_n(z) = \frac{n}{m}\varphi_n(z) + \frac{m-n}{m}\frac{1}{z}$$

together with $n/m \to c$ yields

$$\sup_{z\in I} |\widetilde{\varphi}_n(z) - \widetilde{\varphi}(z)| \to 0 \qquad \text{almost surely.}$$

Since φ_n, $\widetilde{\varphi}_n$, φ, and $\widetilde{\varphi}$ are uniformly bounded on I for large n, the same holds for the products:

$$\sup_{z\in I} |d_n(z) - d(z)| \to 0 \qquad \text{almost surely.}$$

As every subsequence admits an almost-surely convergent further subsequence, the full sequence converges in probability. □

Lemma 2.9 (Continuity sets under weak convergence in probability)**.** *Let ν_n be random probability measures on a metric space E, and let ν be a deterministic probability measure on E. Assume*

$$\nu_n \Rightarrow \nu \qquad \textit{weakly in probability.}$$

Then for every Borel set $A \subset E$ such that $\nu(\partial A) = 0$,

$$\nu_n(A) \to \nu(A) \qquad \textit{in probability.}$$

Proof. Fix $\eta > 0$. By regularity of ν and Urysohn's lemma, there exist bounded continuous functions $f_-, f_+ : E \to [0,1]$ such that

$$f_- \le \mathbf{1}_A \le f_+$$

and

$$\int (f_+ - f_-) \, d\nu < \eta.$$

Weak convergence in probability implies

$$\int f_\pm \, d\nu_n \to \int f_\pm \, d\nu \qquad \text{in probability.}$$

Since

$$\int f_- \, d\nu_n \le \nu_n(A) \le \int f_+ \, d\nu_n,$$

it follows that, with probability tending to one,

$$\nu(A) - 2\eta \le \nu_n(A) \le \nu(A) + 2\eta.$$

Because $\eta > 0$ is arbitrary, the result follows. □

Lemma 2.10 (Continuous pushforward under weak convergence in probability)**.** *Let E and F be metric spaces, let $T : E \to F$ be continuous, let ν_n be random finite Borel measures on E, and let ν be a deterministic finite Borel measure on E. If*

$$\nu_n \Rightarrow \nu \qquad \text{weakly in probability,}$$

then

$$T_\# \nu_n \Rightarrow T_\# \nu \qquad \text{weakly in probability.}$$

Proof. For every bounded continuous $g : F \to \mathbb{R}$, the composition $g \circ T$ is bounded and continuous on E, hence

$$\int g\, d(T_\# \nu_n) = \int g \circ T\, d\nu_n \to \int g \circ T\, d\nu = \int g\, d(T_\# \nu)$$

in probability. □

Lemma 2.11 (Vague convergence plus total masses implies weak convergence)**.** *Let S be a locally compact metric space and let μ_n, μ be finite Borel measures on S. Assume*

$$\mu_n \xrightarrow{v} \mu \qquad \text{vaguely}$$

and

$$\mu_n(S) \to \mu(S).$$

Then $\mu_n \Rightarrow \mu$ weakly as finite measures on S.

Proof. Let $f : S \to \mathbb{R}$ be bounded and continuous, and fix $\eta > 0$. By inner regularity of μ, choose $\chi \in C_c(S)$ with $0 \le \chi \le 1$ such that

$$\int (1 - \chi)\, d\mu < \eta.$$

Then $f\chi \in C_c(S)$, so vague convergence gives

$$\int f\chi\, d\mu_n \to \int f\chi\, d\mu.$$

Also, because $\chi \in C_c(S)$,

$$\int \chi\, d\mu_n \to \int \chi\, d\mu,$$

and therefore

$$\int (1 - \chi)\, d\mu_n = \mu_n(S) - \int \chi\, d\mu_n \to \mu(S) - \int \chi\, d\mu = \int (1 - \chi)\, d\mu.$$

Hence, for all large n,

$$\int (1 - \chi)\, d\mu_n \le \eta + \int (1 - \chi)\, d\mu \le 2\eta.$$

It follows that

$$\left| \int f\, d\mu_n - \int f\, d\mu \right| \le \left| \int f\chi\, d\mu_n - \int f\chi\, d\mu \right| + \|f\|_\infty \int (1 - \chi)\, d\mu_n + \|f\|_\infty \int (1 - \chi)\, d\mu.$$

Taking the limsup and using the preceding bounds yields

$$\limsup_{n \to \infty} \left| \int f\, d\mu_n - \int f\, d\mu \right| \le 3\, \|f\|_\infty\, \eta.$$

Since $\eta > 0$ is arbitrary, $\int f\, d\mu_n \to \int f\, d\mu$ for every bounded continuous f, which is the desired weak convergence. □

Lemma 2.12 (Vague convergence plus total masses in probability implies weak convergence)**.** *Let S be a locally compact metric space, let μ_n be random finite Borel measures on S, and let μ be a deterministic finite measure on S. Assume that*

$$\mu_n \xrightarrow{v} \mu \qquad \textit{vaguely in probability}$$

and

$$\mu_n(S) \to \mu(S) \qquad \textit{in probability.}$$

Then $\mu_n \Rightarrow \mu$ weakly in probability as finite measures on S.

Proof. Let (μ_{n_k}) be an arbitrary subsequence. By the standard subsequence principle for convergence in probability, there exists a further subsequence, still denoted (μ_{n_k}) for simplicity, such that

$$\mu_{n_k} \xrightarrow{v} \mu \qquad \text{vaguely almost surely}$$

and

$$\mu_{n_k}(S) \to \mu(S) \qquad \text{almost surely.}$$

On that almost-sure event, Lemma 2.11 yields

$$\mu_{n_k} \Rightarrow \mu \qquad \text{weakly.}$$

Since every subsequence admits a further subsequence along which weak convergence holds almost surely, the full sequence converges weakly in probability. □

2.3 The model-dependent input

For $z > \|X_n\|_{\mathrm{op}}$, define

$$\begin{aligned}
\mathcal{A}_n(z) &:= U_n^* \Big[z(z^2 \mathrm{I}_n - X_n X_n^*)^{-1} \Big] U_n, \\
\mathcal{D}_n(z) &:= V_n^* \Big[z(z^2 \mathrm{I}_m - X_n^* X_n)^{-1} \Big] V_n, \\
\mathcal{B}_n(z) &:= U_n^* \Big[X_n(z^2 \mathrm{I}_m - X_n^* X_n)^{-1} \Big] V_n.
\end{aligned}$$

Assumption 2.13 (Uniform resolvent-compression hypothesis)**.** For every compact interval $I \subset\subset (b, \infty)$ there exists a deterministic sequence $\beta_n(I) \to 0$ such that

$$\mathbb{P}\left(\begin{array}{c} \|X_n\|_{\mathrm{op}} < \inf I, \\ \sup_{z \in I} \|\mathcal{A}_n(z) - \varphi_n(z) \mathrm{I}_{r_n}\|_{\mathrm{op}} + \sup_{z \in I} \|\mathcal{D}_n(z) - \widetilde{\varphi}_n(z) \mathrm{I}_{r_n}\|_{\mathrm{op}} + \sup_{z \in I} \|\mathcal{B}_n(z)\|_{\mathrm{op}} \leq \beta_n(I) \end{array} \right) \to 1.$$

Remark 2.14. Assumption 2.13 is exactly the point at which broader random-matrix inputs can enter. For Haar-singular-vector ensembles we verify it directly below. For more general i.i.d. or correlated rectangular models, one would expect such bounds to follow from suitable isotropic or anisotropic local laws and bilinear-form resolvent estimates; see [11, 4, 6, 13, 9]. We do not pursue that verification here.

3 Preliminaries

3.1 Rank inequalities

Theorem 3.1 (Rank inequality for Hermitian matrices)**.** *Let A, B be $N \times N$ Hermitian matrices and set $k := \operatorname{rank}(A - B)$. Then for every $x \in \mathbb{R}$,*

$$|F^A(x) - F^B(x)| \leq \frac{k}{N}.$$

Proof. This is the standard interlacing consequence of the min–max principle. □

Corollary 3.2 (Rank inequality for Gram matrices)**.** *Let $Q, \widetilde{Q} \in \mathbb{F}^{n\times m}$ and set $k := \operatorname{rank}(Q - \widetilde{Q})$. Then*

$$\|F^{QQ^*} - F^{\widetilde{Q}\widetilde{Q}^*}\|_\infty \leq \frac{2k}{n}, \qquad \|F^{Q^*Q} - F^{\widetilde{Q}^*\widetilde{Q}}\|_\infty \leq \frac{2k}{m}.$$

Proof. Since

$$QQ^* - \widetilde{Q}\widetilde{Q}^* = (Q - \widetilde{Q})Q^* + \widetilde{Q}(Q^* - \widetilde{Q}^*),$$

the rank of the difference is at most $2k$. Apply Theorem 3.1. □

3.2 Linearization and the determinant equation

Lemma 3.3 (Linearization)**.** *For $M \in \mathbb{F}^{n\times m}$ define*

$$\mathcal{M} := \begin{pmatrix} 0 & M \\ M^* & 0 \end{pmatrix} \in \mathbb{F}^{(n+m)\times(n+m)}.$$

The multiset of positive eigenvalues of $\mathcal{M}$ is exactly the multiset of positive singular values of M. Equivalently, if $q = \operatorname{rank}(M)$ and the positive singular values are $\sigma_1(M) \geq \cdots \geq \sigma_q(M) > 0$, then the nonzero eigenvalues of $\mathcal{M}$ are

$$\sigma_1(M), \ldots, \sigma_q(M), -\sigma_q(M), \ldots, -\sigma_1(M),$$

counted with multiplicity; all remaining eigenvalues are zero.

Proof. Let $M = U\Sigma V^*$ be a singular-value decomposition with positive singular values $\sigma_1, \ldots, \sigma_q$ and associated singular vectors u_j, v_j. For each $1 \leq j \leq q$,

$$\mathcal{M}\begin{pmatrix} u_j \\ v_j \end{pmatrix} = \sigma_j \begin{pmatrix} u_j \\ v_j \end{pmatrix}, \qquad \mathcal{M}\begin{pmatrix} u_j \\ -v_j \end{pmatrix} = -\sigma_j \begin{pmatrix} u_j \\ -v_j \end{pmatrix}.$$

Vectors in the left and right null spaces give the remaining zero eigenvalues. This proves the claim. □

Lemma 3.4 (Low-rank determinant equation)**.** *Let $\mathcal{X}$ be Hermitian and let $\mathcal{P} = E\mathcal{K}E^*$ be Hermitian, where $E \in \mathbb{F}^{N\times K}$ has full column rank and $\mathcal{K} \in \mathbb{F}^{K\times K}$ is Hermitian. For $z \notin \operatorname{spec}(\mathcal{X})$ let $\mathcal{G}(z) = (z\mathrm{I} - \mathcal{X})^{-1}$. Then*

$$\det(z\mathrm{I} - (\mathcal{X} + \mathcal{P})) = \det(z\mathrm{I} - \mathcal{X})\,\det(\mathrm{I}_K - \mathcal{K}E^*\mathcal{G}(z)E).$$

Hence, for $z \notin \operatorname{spec}(\mathcal{X})$,

$$z \in \operatorname{spec}(\mathcal{X} + \mathcal{P}) \iff \det(\mathrm{I}_K - \mathcal{K}E^*\mathcal{G}(z)E) = 0.$$

Proof. This is the matrix determinant lemma:

$$z\mathrm{I} - (\mathcal{X} + \mathcal{P}) = (z\mathrm{I} - \mathcal{X})\big(\mathrm{I} - \mathcal{G}(z)E\mathcal{K}E^*\big),$$

so

$$\det(z\mathrm{I} - (\mathcal{X} + \mathcal{P})) = \det(z\mathrm{I} - \mathcal{X})\,\det(\mathrm{I} - \mathcal{G}(z)E\mathcal{K}E^*).$$

Since $\det(\mathrm{I} - AB) = \det(\mathrm{I} - BA)$ whenever the products are well defined,

$$\det(\mathrm{I} - \mathcal{G}(z)E\mathcal{K}E^*) = \det(\mathrm{I}_K - \mathcal{K}E^*\mathcal{G}(z)E).$$

This proves both assertions. □

Lemma 3.5 (Resolvent block formula)**.** *Let $X \in \mathbb{F}^{n\times m}$ and*

$$\mathcal{X} = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}.$$

If $z > \|X\|_{\mathrm{op}}$, then

$$(z\mathrm{I} - \mathcal{X})^{-1} = \begin{pmatrix} z(z^2\mathrm{I}_n - XX^*)^{-1} & X(z^2\mathrm{I}_m - X^*X)^{-1} \\ X^*(z^2\mathrm{I}_n - XX^*)^{-1} & z(z^2\mathrm{I}_m - X^*X)^{-1} \end{pmatrix}.$$

Proof. A direct block multiplication verifies the identity. □

Lemma 3.6 (Weyl's inequality)**.** *If A, B are Hermitian $N \times N$ matrices with ordered eigenvalues $\lambda_1(\cdot) \geq \cdots \geq \lambda_N(\cdot)$, then*

$$|\lambda_i(A) - \lambda_i(B)| \leq \|A - B\|_{\mathrm{op}} \qquad (1 \leq i \leq N).$$

Proof. This is the standard minimax characterization of eigenvalues. □

Lemma 3.7 (Counting identity via inertia)**.** *Let $\mathcal{X}$ be an $N \times N$ Hermitian matrix, let $E \in \mathbb{F}^{N\times K}$ have full column rank, let $\mathcal{K} \in \mathbb{F}^{K\times K}$ be Hermitian, let $\mathcal{P} = E\mathcal{K}E^*$, and set $\mathcal{W} = \mathcal{X} + \mathcal{P}$. For $z > \|\mathcal{X}\|_{\mathrm{op}}$, define*

$$\mathcal{G}(z) := (z\mathrm{I} - \mathcal{X})^{-1}, \qquad G(z) := E^*\mathcal{G}(z)E, \qquad H(z) := G(z)^{1/2}\mathcal{K}G(z)^{1/2}.$$

Then

$$\#\{\lambda \in \mathrm{spec}(\mathcal{W}) : \lambda > z\} = \#\{\lambda \in \mathrm{spec}(H(z)) : \lambda > 1\}.$$

In particular, if $\mathcal{W}$ is the linearization of $W = X + P$, then for $z > \|X\|_{\mathrm{op}}$,

$$\#\{i : \sigma_i(W) > z\} = \#\{\lambda \in \mathrm{spec}(H(z)) : \lambda > 1\}.$$

Proof. Since $z > \|\mathcal{X}\|_{\mathrm{op}}$, the matrix

$$z\mathrm{I} - \mathcal{X}$$

is positive definite. Therefore

$$z\mathrm{I} - \mathcal{W} = (z\mathrm{I} - \mathcal{X})^{1/2}\Big(\mathrm{I} - (z\mathrm{I} - \mathcal{X})^{-1/2}E\mathcal{K}E^*(z\mathrm{I} - \mathcal{X})^{-1/2}\Big)(z\mathrm{I} - \mathcal{X})^{1/2}.$$

Congruence by the positive definite factor $(z\mathrm{I} - \mathcal{X})^{1/2}$ preserves inertia, so the number of negative eigenvalues of $z\mathrm{I} - \mathcal{W}$ equals the number of negative eigenvalues of

$$\mathrm{I} - \widetilde{E}\mathcal{K}\widetilde{E}^*, \qquad \widetilde{E} := (z\mathrm{I} - \mathcal{X})^{-1/2}E.$$

The latter is exactly the number of eigenvalues of $\widetilde{E}\mathcal{K}\widetilde{E}^*$ that exceed 1.

Now $\widetilde{E}\mathcal{K}\widetilde{E}^*$ and $\mathcal{K}\widetilde{E}^*\widetilde{E}$ have the same nonzero eigenvalues, counted with algebraic multiplicity. Since

$$\widetilde{E}^*\widetilde{E} = E^*(z\mathrm{I} - \mathcal{X})^{-1}E = G(z),$$

the nonzero eigenvalues of $\widetilde{E}\mathcal{K}\widetilde{E}^*$ coincide with those of $\mathcal{K}G(z)$. Because $G(z) \succ 0$, the matrix $\mathcal{K}G(z)$ is similar to

$$G(z)^{1/2}\mathcal{K}G(z)^{1/2} = H(z),$$

hence they have the same spectrum. Therefore the number of eigenvalues of $H(z)$ that exceed 1 equals the number of negative eigenvalues of $z\mathrm{I} - \mathcal{W}$. But the latter is precisely the number of eigenvalues of $\mathcal{W}$ that are larger than z. For a singular-value linearization, eigenvalues of $\mathcal{W}$ above the positive level z are precisely singular values of W above z, by Lemma 3.3. □

4 Main results

4.1 Global law

Theorem 4.1 (The global law is unchanged)**.** *Under Assumptions 2.1 and 2.2,*

$$\|F^{W_n^* W_n} - F^{X_n^* X_n}\|_\infty \le \frac{2r_n}{m} \to 0.$$

Consequently, $F^{W_n^* W_n}$ and $F^{X_n^* X_n}$ have the same weak limit.

Proof. Apply Corollary 3.2 with $Q = W_n$ and $\widetilde{Q} = X_n$. □

4.2 Uniform ordered matching away from the threshold

Fix $\delta > 0$ and define

$$k_{n,\delta} := \#\{1 \le i \le r_n : \theta_{n,i} \ge \theta_\star + \delta\}.$$

Set

$$t_\delta := \rho(\theta_\star + \delta) > b, \qquad s_\delta := \frac{b + t_\delta}{2}, \qquad z_\delta := 1 + \max\{t_\delta, \rho(\theta_{\max})\}, \qquad I_\delta := [s_\delta, z_\delta]. \tag{1}$$

The interval I_δ is nonempty and satisfies

$$s_\delta < t_\delta < z_\delta, \qquad \rho(\theta) \in [t_\delta, z_\delta - 1] \quad \text{for every } \theta \in [\theta_\star + \delta, \theta_{\max}].$$

Lemma 4.2 (Deterministic envelope from convergence in probability)**.** *Let $\alpha_n \ge 0$ be random and assume $\alpha_n \to 0$ in probability. Then there exists a deterministic sequence $\zeta_n \downarrow 0$ such that*

$$\mathbb{P}(\alpha_n \le \zeta_n) \to 1.$$

Proof. This is the standard diagonal-envelope construction. Define

$$k_n := \max\Big(\{1\} \cup \big\{k \in \{1,\ldots,n\} : \mathbb{P}(\alpha_n > 1/k) \le 1/k\big\}\Big),$$

so that k_n is well defined for every n. Set

$$\zeta_n^{(0)} := \frac{1}{k_n}, \qquad \zeta_n := \sup_{m \ge n} \zeta_m^{(0)}.$$

Then $\zeta_n \downarrow 0$. Indeed, for every fixed $K \in \mathbb{N}$, convergence $\alpha_n \to 0$ in probability implies

$$\mathbb{P}(\alpha_n > 1/K) \to 0,$$

so for all sufficiently large n one has $\mathbb{P}(\alpha_n > 1/K) \le 1/K$, hence $k_n \ge K$ and therefore $\zeta_n^{(0)} \le 1/K$. Since K is arbitrary, $\zeta_n^{(0)} \to 0$, and thus also $\zeta_n \downarrow 0$. Finally, by construction,

$$\mathbb{P}(\alpha_n \le \zeta_n) \ge \mathbb{P}(\alpha_n \le \zeta_n^{(0)}) \ge 1 - \zeta_n^{(0)} \to 1.$$

□

Theorem 4.3 (Uniform ordered matching)**.** *Assume Assumptions 2.1, 2.2, and 2.13. Fix $\delta > 0$. Define*

$$\alpha_{n,\delta} := \sup_{z \in I_\delta} |d_n(z) - d(z)|.$$

Then $\alpha_{n,\delta} \to 0$ in probability by Proposition 2.8. Let $\zeta_{n,\delta} \downarrow 0$ be deterministic with

$$\mathbb{P}(\alpha_{n,\delta} \le \zeta_{n,\delta}) \to 1.$$

Then there exists $C_\delta > 0$ such that, with probability $1 - o(1)$,

$$\max_{1 \le i \le k_{n,\delta}} |\sigma_i(W_n) - \rho(\theta_{n,i})| \le C_\delta\big(\beta_n(I_\delta) + \zeta_{n,\delta}\big), \tag{2}$$

$$\sigma_{k_{n,\delta}+1}(W_n) \le t_\delta + C_\delta\big(\beta_n(I_\delta) + \zeta_{n,\delta}\big). \tag{3}$$

The first estimate is vacuous if $k_{n,\delta} = 0$. Consequently,

$$\max_{1 \le i \le k_{n,\delta}} |\lambda_i(W_n^* W_n) - \Lambda(\theta_{n,i})| \le C_\delta\big(\beta_n(I_\delta) + \zeta_{n,\delta}\big).$$

If, in addition, $0 < \delta < \theta_\star$ and $\theta_{n,1} \le \theta_\star - \delta$, then for every fixed $\varepsilon > 0$,

$$\mathbb{P}(\|W_n\|_{\mathrm{op}} \le b + \varepsilon) \to 1.$$

Remark 4.4. The theorem is entirely deterministic once the two inputs

$$\alpha_{n,\delta} = \sup_{z \in I_\delta} |d_n(z) - d(z)| \qquad \text{and} \qquad \beta_n(I_\delta)$$

are available. For Haar-singular-vector ensembles proved in Section 7,

$$\beta_n(I_\delta) \lesssim \sqrt{\frac{r_n}{n}} + \sqrt{\frac{\log n}{n}}.$$

Remark 4.5. No spacing assumption among the supercritical spikes is required. In particular, clusters of spikes whose images under ρ are much closer than the eventual error scale are still matched in order.

4.3 The outlier point measure above a fixed separation

Fix $\varepsilon > 0$ and define

$$\gamma_\varepsilon := (b+\varepsilon)^2.$$

Let

$$\mu_{n,\varepsilon} := \frac{1}{r_n} \sum_{j=1}^{m} \mathbf{1}\{\lambda_j(W_n^* W_n) > \gamma_\varepsilon\}\, \delta_{\lambda_j(W_n^* W_n)}.$$

Theorem 4.6 (Vague convergence of the outlier measure)**.** *Assume the hypotheses of Theorem 4.3, and suppose further that*

$$H_n \Rightarrow H$$

weakly in probability for some probability measure H on $(0,\infty)$. (If the spike is deterministic, this is just ordinary weak convergence.) Then for every fixed $\varepsilon > 0$ and every test function $f \in C_c((\gamma_\varepsilon,\infty))$,

$$\int f\, d\mu_{n,\varepsilon} \longrightarrow \int f\, d\mu_\varepsilon \qquad \textit{in probability,}$$

where

$$\mu_\varepsilon := (\Lambda_\# H)|_{(\gamma_\varepsilon,\infty)}.$$

Equivalently, $\mu_{n,\varepsilon}$ converges vaguely in probability *on $(\gamma_\varepsilon,\infty)$ to μ_ε.*

Corollary 4.7 (Singular-value-scale outlier law)**.** *Under the hypotheses of Theorem 4.6, define*

$$\nu_{n,\varepsilon} := \frac{1}{r_n} \sum_{j=1}^{n} \mathbf{1}\{\sigma_j(W_n) > b+\varepsilon\}\, \delta_{\sigma_j(W_n)}.$$

Then $\nu_{n,\varepsilon}$ converges vaguely in probability on $(b+\varepsilon,\infty)$ to

$$\nu_\varepsilon := (\rho_\# H)|_{(b+\varepsilon,\infty)}.$$

If, in addition,

$$H\big(\{\theta : \rho(\theta) = b+\varepsilon\}\big) = 0,$$

then $\nu_{n,\varepsilon}$ converges weakly in probability as a finite measure to ν_ε.

Proof. Let $T : (\gamma_\varepsilon,\infty) \to (b+\varepsilon,\infty)$ be given by $T(x) = \sqrt{x}$. Since T is a homeomorphism,

$$\nu_{n,\varepsilon} = T_\# \mu_{n,\varepsilon}, \qquad \nu_\varepsilon = T_\# \mu_\varepsilon,$$

where $\mu_\varepsilon = (\Lambda_\# H)|_{(\gamma_\varepsilon,\infty)}$.

For the vague-convergence statement, let $g \in C_c((b+\varepsilon,\infty))$. Then $g \circ T \in C_c((\gamma_\varepsilon,\infty))$, so Theorem 4.6 gives

$$\int g\, d\nu_{n,\varepsilon} = \int g \circ T\, d\mu_{n,\varepsilon} \to \int g \circ T\, d\mu_\varepsilon = \int g\, d\nu_\varepsilon \qquad \text{in probability.}$$

Thus $\nu_{n,\varepsilon}$ converges vaguely in probability to ν_ε on $(b+\varepsilon,\infty)$.

If, in addition,

$$H\big(\{\theta : \rho(\theta) = b+\varepsilon\}\big) = 0,$$

then

$$(\rho_\# H)(\{b+\varepsilon\}) = 0, \qquad \text{equivalently} \qquad (\Lambda_\# H)(\{\gamma_\varepsilon\}) = 0.$$

By Corollary 4.8, $\mu_{n,\varepsilon}$ converges weakly in probability as a finite measure to μ_ε. Since Lemma 2.10 now applies to finite measures, it follows that

$$\nu_{n,\varepsilon} = T_{\#}\mu_{n,\varepsilon} \Rightarrow T_{\#}\mu_\varepsilon = \nu_\varepsilon$$

weakly in probability. $\square$

Corollary 4.8 (Upgrade under no boundary mass). *Under the hypotheses of Theorem 4.6, assume additionally that*

$$H\big(\{\theta : \Lambda(\theta) = \gamma_\varepsilon\}\big) = 0.$$

Then

$$\frac{1}{r_n}\#\{j : \lambda_j(W_n^* W_n) > \gamma_\varepsilon\} \to H\big(\{\theta : \Lambda(\theta) > \gamma_\varepsilon\}\big) \qquad \text{in probability.}$$

Equivalently,

$$\frac{1}{r_n}\#\{j : \sigma_j(W_n) > b+\varepsilon\} \to H\big(\{\theta : \rho(\theta) > b+\varepsilon\}\big) \qquad \text{in probability.}$$

Hence $\mu_{n,\varepsilon}$ converges weakly in probability as a finite measure to μ_ε. If the limiting mass is positive, then

$$\mathbb{P}\big(\mu_{n,\varepsilon}\big((\gamma_\varepsilon,\infty)\big) > 0\big) \to 1,$$

and the normalized outlier law

$$\widehat{\mu}_{n,\varepsilon} := \begin{cases} \dfrac{\mu_{n,\varepsilon}}{\mu_{n,\varepsilon}((\gamma_\varepsilon,\infty))}, & \mu_{n,\varepsilon}((\gamma_\varepsilon,\infty)) > 0, \\ 0, & \mu_{n,\varepsilon}((\gamma_\varepsilon,\infty)) = 0, \end{cases}$$

converges weakly in probability to the normalized restriction of $\Lambda_{\#}H$ to $(\gamma_\varepsilon,\infty)$.

Remark 4.9. The additional boundary-mass hypothesis in Corollary 4.8 is essential if one wants convergence of the total mass of $\mu_{n,\varepsilon}$, i.e. convergence of the normalized number of outliers above the strict threshold. Without it, Theorem 4.6 still gives the correct vague limit against test functions supported a positive distance above γ_ε, but the total mass need not converge.

5 Proof of Theorem 4.3

5.1 Linearized determinant equation

Define the linearizations

$$\mathcal{X}_n := \begin{pmatrix} 0 & X_n \\ X_n^* & 0 \end{pmatrix}, \qquad \mathcal{P}_n := \begin{pmatrix} 0 & P_n \\ P_n^* & 0 \end{pmatrix}, \qquad \mathcal{W}_n := \mathcal{X}_n + \mathcal{P}_n.$$

Write the columns of U_n and V_n as $u_{1,n},\dots,u_{r_n,n}$ and $v_{1,n},\dots,v_{r_n,n}$. Set

$$e_i := \begin{pmatrix} u_{i,n} \\ 0 \end{pmatrix}, \qquad f_i := \begin{pmatrix} 0 \\ v_{i,n} \end{pmatrix}, \qquad E_n := [e_1,\dots,e_{r_n}, f_1,\dots,f_{r_n}],$$

and

$$\mathcal{K}_n := \begin{pmatrix} 0 & \Theta_n \\ \Theta_n & 0 \end{pmatrix}.$$

Then $\mathcal{P}_n = E_n \mathcal{K}_n E_n^*$.

For $z > \|X_n\|_{\mathrm{op}}$, define

$$G_n(z) := E_n^*(z\mathrm{I} - \mathcal{X}_n)^{-1} E_n.$$

By Lemma 3.5,

$$G_n(z) = \begin{pmatrix} \mathcal{A}_n(z) & \mathcal{B}_n(z) \\ \mathcal{B}_n(z)^* & \mathcal{D}_n(z) \end{pmatrix}.$$

By Lemma 3.4,

$$z \in \mathrm{spec}(\mathcal{W}_n) \iff \det(\mathrm{I}_{2r_n} - \mathcal{K}_n G_n(z)) = 0. \tag{4}$$

By Lemma 3.3, eigenvalues of $\mathcal{W}_n$ above any positive level are precisely the singular values of W_n above that level.

5.2 Comparison with the scalar model

Define the deterministic block approximation

$$G_n^{(0)}(z) := \begin{pmatrix} \varphi_n(z)\mathrm{I}_{r_n} & 0 \\ 0 & \widetilde{\varphi}_n(z)\mathrm{I}_{r_n} \end{pmatrix}.$$

On the compression event from Assumption 2.13,

$$\sup_{z \in I_\delta} \left\| G_n(z) - G_n^{(0)}(z) \right\|_{\mathrm{op}} \le 4\,\beta_n(I_\delta).$$

Indeed, the operator norm of a 2×2 block matrix is bounded by the sum of the operator norms of its four blocks.

Define

$$H_n(z) := G_n(z)^{1/2} \mathcal{K}_n G_n(z)^{1/2}, \qquad H_n^{(0)}(z) := G_n^{(0)}(z)^{1/2} \mathcal{K}_n G_n^{(0)}(z)^{1/2}.$$

Since $G_n^{(0)}(z)$ is block diagonal,

$$H_n^{(0)}(z) = \sqrt{d_n(z)} \begin{pmatrix} 0 & \Theta_n \\ \Theta_n & 0 \end{pmatrix},$$

whose eigenvalues are

$$\pm\theta_{n,1}\sqrt{d_n(z)}, \dots, \pm\theta_{n,r_n}\sqrt{d_n(z)}.$$

Lemma 5.1 (Square-root perturbation)**.** *Let A, B be positive definite and assume $\lambda_{\min}(A) \ge \alpha > 0$ and $\|A - B\|_{\mathrm{op}} \le \alpha/2$. Then*

$$\left\| A^{1/2} - B^{1/2} \right\|_{\mathrm{op}} \le \frac{\|A - B\|_{\mathrm{op}}}{\sqrt{\alpha}}.$$

Proof. By Weyl's inequality, $B \ge (\alpha/2)\mathrm{I}$. We use the integral representation, valid for every positive definite matrix T,

$$T^{1/2} = \frac{1}{\pi} \int_0^\infty t^{-1/2}\, T(T + t\mathrm{I})^{-1}\, dt,$$

where the integral converges in operator norm; the formula follows from the scalar identity by the spectral theorem. Thus

$$A^{1/2} - B^{1/2} = \frac{1}{\pi} \int_0^\infty t^{1/2}(A + t\mathrm{I})^{-1}(A - B)(B + t\mathrm{I})^{-1}\, dt.$$

Since $A, B \ge (\alpha/2)\mathrm{I}$,

$$\left\|A^{1/2} - B^{1/2}\right\|_{\mathrm{op}} \le \frac{\|A-B\|_{\mathrm{op}}}{\pi} \int_0^\infty \frac{t^{1/2}}{(t+\alpha/2)^2}\,dt.$$

The change of variables $t = (\alpha/2)u$ and the beta-integral $\int_0^\infty u^{1/2}(1+u)^{-2}\,du = \pi/2$ give

$$\int_0^\infty \frac{t^{1/2}}{(t+\alpha/2)^2}\,dt = \frac{\pi}{\sqrt{2\alpha}}.$$

Therefore

$$\left\|A^{1/2} - B^{1/2}\right\|_{\mathrm{op}} \le \frac{\|A-B\|_{\mathrm{op}}}{\sqrt{2\alpha}} \le \frac{\|A-B\|_{\mathrm{op}}}{\sqrt{\alpha}}.$$

□

Lemma 5.2 (Uniform control of $H_n - H_n^{(0)}$ on compact intervals)**.** *Let $I = [z_-, z_+] \subset\subset (b, \infty)$ and set*

$$\bar{b}_I := b + \frac{z_- - b}{2}.$$

Then there exist a deterministic constant C_I and an integer $n_0(I)$ such that, for every $n \ge n_0(I)$, on the event

$$\|X_n\|_{\mathrm{op}} \le \bar{b}_I$$

and the compression event from Assumption 2.13 on I, one has

$$\sup_{z\in I} \left\|H_n(z) - H_n^{(0)}(z)\right\|_{\mathrm{op}} \le C_I\,\beta_n(I).$$

Proof. For every $z \in I$,

$$\varphi_n(z) \ge \frac{1}{z} \ge \frac{1}{z_+}, \qquad \widetilde{\varphi}_n(z) \ge \frac{1}{z} \ge \frac{1}{z_+},$$

so

$$\lambda_{\min}(G_n^{(0)}(z)) \ge \frac{1}{z_+}.$$

Because $\beta_n(I) \to 0$, there exists $n_0(I)$ such that for all $n \ge n_0(I)$ the compression event implies

$$\left\|G_n(z) - G_n^{(0)}(z)\right\|_{\mathrm{op}} \le \frac{1}{2z_+} \qquad (z \in I).$$

Hence Weyl gives

$$\lambda_{\min}(G_n(z)) \ge \frac{1}{2z_+} \qquad (z \in I).$$

Applying Lemma 5.1 with $A = G_n^{(0)}(z)$ and $B = G_n(z)$ yields

$$\sup_{z\in I} \left\|G_n(z)^{1/2} - G_n^{(0)}(z)^{1/2}\right\|_{\mathrm{op}} \le C_I\,\beta_n(I).$$

Moreover, because E_n has orthonormal columns,

$$\|G_n(z)\|_{\mathrm{op}} \le \left\|(z\mathrm{I} - \mathcal{X}_n)^{-1}\right\|_{\mathrm{op}} = \frac{1}{z - \|X_n\|_{\mathrm{op}}} \le \frac{2}{z_- - b}$$

on the event $\{\|X_n\|_{\mathrm{op}} \le \bar{b}_I\}$. Likewise $\left\|G_n^{(0)}(z)\right\|_{\mathrm{op}}$ is bounded uniformly on I by a deterministic constant depending only on I. Since $\|\mathcal{K}_n\|_{\mathrm{op}} \le \theta_{\max}$, we may write

$$H_n - H_n^{(0)} = (G_n^{1/2} - G_n^{(0)1/2})\mathcal{K}_n G_n^{1/2} + G_n^{(0)1/2}\mathcal{K}_n(G_n^{1/2} - G_n^{(0)1/2}),$$

and each term is bounded by $C_I\beta_n(I)$ uniformly in $z \in I$. □

5.3 Scalar crossings

For each spike index i, define

$$g_{n,i}(z) := \theta_{n,i}\sqrt{d_n(z)}, \qquad z > \|X_n\|_{\mathrm{op}}\,.$$

These functions are strictly decreasing by Lemma 2.5.

Lemma 5.3 (Scalar crossings)**.** *Fix $\delta > 0$. With probability $1 - o(1)$, for every $i \le k_{n,\delta}$ the equation*

$$g_{n,i}(z) = 1$$

has a unique solution $z_{n,i}^{(0)} \in I_\delta$. Moreover, the map $i \mapsto z_{n,i}^{(0)}$ is nonincreasing.

Proof. If $k_{n,\delta} = 0$, there is nothing to prove, so assume $k_{n,\delta} \ge 1$.

Since $s_\delta < t_\delta = \rho(\theta_\star + \delta)$ and d is strictly decreasing,

$$d(s_\delta) > d(t_\delta) = \frac{1}{(\theta_\star + \delta)^2}.$$

By Proposition 2.8,

$$d_n(s_\delta) > \frac{1}{(\theta_\star + \delta)^2}$$

with probability $1 - o(1)$, and therefore

$$g_{n,i}(s_\delta) > 1 \qquad (1 \le i \le k_{n,\delta})$$

on the same event.

We next show that

$$d(z_\delta) < \frac{1}{\theta_{\max}^2}.$$

If $\theta_{\max} > \theta_\star$, then $\rho(\theta_{\max}) > b$ and

$$d(z_\delta) < d(\rho(\theta_{\max})) = \frac{1}{\theta_{\max}^2}$$

because $z_\delta > \rho(\theta_{\max})$. If $\theta_{\max} \le \theta_\star$, then $\theta_{\max} \le \theta_\star + \delta$ and $z_\delta > t_\delta$, so

$$d(z_\delta) < d(t_\delta) = \frac{1}{(\theta_\star + \delta)^2} \le \frac{1}{\theta_{\max}^2}.$$

Thus in all cases $d(z_\delta) < 1/\theta_{\max}^2$. A second application of Proposition 2.8 gives

$$d_n(z_\delta) < \frac{1}{\theta_{\max}^2}$$

with probability $1 - o(1)$, hence

$$g_{n,i}(z_\delta) < 1 \qquad (1 \le i \le r_n)$$

on the same event.

For every $i \le k_{n,\delta}$, continuity and strict monotonicity of $g_{n,i}$ therefore give a unique solution $z_{n,i}^{(0)} \in (s_\delta, z_\delta)$ of $g_{n,i}(z) = 1$. Finally, if $i < j \le k_{n,\delta}$ then $\theta_{n,i} \ge \theta_{n,j}$, so

$$g_{n,i}(z) \ge g_{n,j}(z) \qquad (z > \|X_n\|_{\mathrm{op}}),$$

which implies $z_{n,i}^{(0)} \ge z_{n,j}^{(0)}$. $\square$

Lemma 5.4 (Deterministic inversion stability)**.** *Fix $\delta > 0$ and let I_δ be defined in* (1)*. There exists $c_\delta^{(d)} > 0$ such that*

$$-d'(z) \geq c_\delta^{(d)} \qquad (z \in I_\delta).$$

Consequently, on the event $\alpha_{n,\delta} \leq \alpha$, one has for every $i \leq k_{n,\delta}$,

$$|z_{n,i}^{(0)} - \rho(\theta_{n,i})| \leq \frac{\alpha}{c_\delta^{(d)}}.$$

Proof. By Lemma 2.5, the deterministic transform satisfies

$$-d'(z) \geq \frac{2}{z^3} \geq \frac{2}{z_\delta^3} \qquad (z \in I_\delta).$$

Set

$$c_\delta^{(d)} := \frac{2}{z_\delta^3} > 0.$$

For $i \leq k_{n,\delta}$ we have $\theta_{n,i} \in [\theta_\star + \delta, \theta_{\max}]$, hence

$$\rho(\theta_{n,i}) \in [t_\delta, z_\delta - 1] \subset I_\delta.$$

Also $z_{n,i}^{(0)} \in I_\delta$ by Lemma 5.3. Since

$$d(\rho(\theta_{n,i})) = \frac{1}{\theta_{n,i}^2} = d_n(z_{n,i}^{(0)}),$$

the event $\{\alpha_{n,\delta} \leq \alpha\}$ implies

$$|d(z_{n,i}^{(0)}) - d(\rho(\theta_{n,i}))| = |d(z_{n,i}^{(0)}) - d_n(z_{n,i}^{(0)})| \leq \alpha.$$

The mean value theorem on I_δ yields the claim. □

Lemma 5.5 (A uniform slope bound)**.** *Fix $\delta > 0$. On the event*

$$\|X_n\|_{\mathrm{op}} \leq b + \frac{t_\delta - b}{4}$$

there exists a deterministic constant $c_\delta^{(g)} > 0$ such that for every $i \leq k_{n,\delta}$,

$$g_{n,i}'(z) \leq -c_\delta^{(g)} \qquad (z \in I_\delta).$$

The same bound holds for the envelope

$$g_{n,\mathrm{env}}(z) := (\theta_\star + \delta)\sqrt{d_n(z)}.$$

Proof. By Lemma 2.5,

$$-d_n'(z) \geq \frac{2}{z_\delta^3} \qquad (z \in I_\delta).$$

Set

$$\bar{b}_\delta := b + \frac{t_\delta - b}{4}, \qquad M_\delta := \frac{z_\delta}{s_\delta^2 - \bar{b}_\delta^2}.$$

On the stated event, every singular value s of X_n satisfies $s \le \bar{b}_\delta$, hence for $z \in I_\delta$,

$$\frac{z}{z^2 - s^2} \le M_\delta.$$

Therefore

$$d_n(z) \le M_\delta^2 \qquad (z \in I_\delta).$$

If $i \le k_{n,\delta}$, then $\theta_{n,i} \ge \theta_\star + \delta$, and consequently

$$g'_{n,i}(z) = \theta_{n,i} \frac{d'_n(z)}{2\sqrt{d_n(z)}} \le -(\theta_\star + \delta) \frac{1}{M_\delta z_\delta^3}.$$

The same computation with $\theta_{n,i}$ replaced by $\theta_\star + \delta$ yields the bound for $g_{n,\mathrm{env}}$. □

5.4 Ordered identification

Let $\lambda_1(H_n(z)) \ge \cdots \ge \lambda_{2r_n}(H_n(z))$ be the ordered eigenvalues. For $1 \le i \le r_n$,

$$\lambda_i(H_n^{(0)}(z)) = g_{n,i}(z).$$

Hence Lemma 3.6 and Lemma 5.2 imply that on the high-probability event from the previous lemmas,

$$\sup_{z \in I_\delta} \max_{1 \le i \le r_n} |\lambda_i(H_n(z)) - g_{n,i}(z)| \le \widetilde{\epsilon}_n, \qquad \widetilde{\epsilon}_n := C_\delta\, \beta_n(I_\delta). \tag{5}$$

Lemma 5.6 (Ordered identification)**.** *With probability* $1 - o(1)$*,*

$$\max_{1 \le i \le k_{n,\delta}} |\sigma_i(W_n) - z_{n,i}^{(0)}| \le \frac{2\widetilde{\epsilon}_n}{c_\delta^{(g)}}.$$

Proof. If $k_{n,\delta} = 0$, the claim is vacuous. Fix $i \le k_{n,\delta}$ and set

$$\Delta := \frac{2\widetilde{\epsilon}_n}{c_\delta^{(g)}}.$$

By Lemma 5.4, $z_{n,i}^{(0)}$ lies within $o(1)$ of $\rho(\theta_{n,i}) \in [t_\delta, z_\delta - 1]$, so for all large n,

$$z_{n,i}^{(0)} \pm \Delta \in I_\delta.$$

Since $g_{n,i}(z_{n,i}^{(0)}) = 1$ and $g'_{n,i} \le -c_\delta^{(g)}$ on I_δ,

$$g_{n,i}(z_{n,i}^{(0)} - \Delta) \ge 1 + 2\widetilde{\epsilon}_n, \qquad g_{n,i}(z_{n,i}^{(0)} + \Delta) \le 1 - 2\widetilde{\epsilon}_n.$$

Using (5),

$$\lambda_i(H_n(z_{n,i}^{(0)} - \Delta)) > 1, \qquad \lambda_i(H_n(z_{n,i}^{(0)} + \Delta)) < 1.$$

By Lemma 3.7, there are at least i singular values of W_n above $z_{n,i}^{(0)} - \Delta$, and at most $i - 1$ singular values above $z_{n,i}^{(0)} + \Delta$. Therefore

$$z_{n,i}^{(0)} - \Delta < \sigma_i(W_n) \le z_{n,i}^{(0)} + \Delta.$$

□

5.5 No extra outliers

Lemma 5.7 (No extra outliers above the δ-supercritical scale)**.** *With probability* $1-o(1)$,

$$\sigma_{k_{n,\delta}+1}(W_n) \le t_\delta + C_\delta\big(\beta_n(I_\delta) + \zeta_{n,\delta}\big).$$

Proof. Define the envelope

$$g_{n,\mathrm{env}}(z) := (\theta_\star + \delta)\sqrt{d_n(z)}.$$

On the event $\alpha_{n,\delta} \le \zeta_{n,\delta}$,

$$g_{n,\mathrm{env}}(t_\delta) \le (\theta_\star + \delta)\sqrt{d(t_\delta) + \zeta_{n,\delta}} = \sqrt{1 + (\theta_\star+\delta)^2\zeta_{n,\delta}} \le 1 + (\theta_\star+\delta)^2\zeta_{n,\delta},$$

where we used $\sqrt{1+x} \le 1 + x$ for $x \ge 0$. Let

$$\Delta_n := \frac{2\widetilde{\epsilon}_n + (\theta_\star+\delta)^2\zeta_{n,\delta}}{c_\delta^{(g)}}.$$

Since $\Delta_n \to 0$, we have $t_\delta + \Delta_n \in I_\delta$ for all large n. By Lemma 5.5,

$$g_{n,\mathrm{env}}(t_\delta + \Delta_n) \le 1 - 2\widetilde{\epsilon}_n.$$

For every index j with $k_{n,\delta} < j \le r_n$, we have $\theta_{n,j} < \theta_\star + \delta$, so

$$g_{n,j}(t_\delta + \Delta_n) \le g_{n,\mathrm{env}}(t_\delta + \Delta_n) \le 1 - 2\widetilde{\epsilon}_n.$$

Hence (5) gives

$$\lambda_j(H_n(t_\delta + \Delta_n)) < 1 \qquad (k_{n,\delta} < j \le r_n).$$

If $j > r_n$, then $\lambda_j(H_n^{(0)}(t_\delta + \Delta_n)) \le 0$, so Weyl's inequality implies

$$\lambda_j(H_n(t_\delta + \Delta_n)) \le \widetilde{\epsilon}_n < 1 \qquad (j > r_n)$$

for all large n. Therefore at most $k_{n,\delta}$ eigenvalues of $H_n(t_\delta + \Delta_n)$ exceed 1. By Lemma 3.7, at most $k_{n,\delta}$ singular values of W_n exceed $t_\delta + \Delta_n$, i.e.

$$\sigma_{k_{n,\delta}+1}(W_n) \le t_\delta + \Delta_n.$$

This is exactly the claimed bound. □

5.6 Conclusion of the proof

Proof of Theorem 4.3. By Lemma 5.6 and Lemma 5.4,

$$|\sigma_i(W_n) - \rho(\theta_{n,i})| \le |\sigma_i(W_n) - z_{n,i}^{(0)}| + |z_{n,i}^{(0)} - \rho(\theta_{n,i})| \le C_\delta\big(\beta_n(I_\delta) + \zeta_{n,\delta}\big)$$

uniformly for $i \le k_{n,\delta}$. This proves (2). The eigenvalue version follows from

$$|\sigma_i(W_n)^2 - \rho(\theta_{n,i})^2| = |\sigma_i(W_n) - \rho(\theta_{n,i})|\,\big(\sigma_i(W_n) + \rho(\theta_{n,i})\big) \le 2(z_\delta + 1)\,|\sigma_i(W_n) - \rho(\theta_{n,i})|$$

for all large n: indeed, $\rho(\theta_{n,i}) \le z_\delta$, while Lemma 5.6 and Lemma 5.4 imply $\sigma_i(W_n) \le z_\delta + 1$ on the relevant high-probability event. The bound (3) is exactly Lemma 5.7.

For the subcritical regime, assume $0 < \delta < \theta_\star$ and $\theta_{n,1} \le \theta_\star - \delta$, and fix $\varepsilon > 0$. Set

$$z_\varepsilon := b + \varepsilon, \qquad J_\varepsilon := [z_\varepsilon, z_\varepsilon + 1].$$

Since d is strictly decreasing on (b, ∞),

$$(\theta_\star - \delta)^2 d(z_\varepsilon) < (\theta_\star - \delta)^2 d(b^+) < 1.$$

Choose $\eta_{\delta,\varepsilon} \in (0, 1/3)$ such that

$$(\theta_\star - \delta)^2 d(z_\varepsilon) \le 1 - 4\eta_{\delta,\varepsilon}.$$

On the event

$$|d_n(z_\varepsilon) - d(z_\varepsilon)| \le \frac{\eta_{\delta,\varepsilon}}{(\theta_\star - \delta)^2},$$

we have

$$(\theta_\star - \delta)^2 d_n(z_\varepsilon) \le 1 - 3\eta_{\delta,\varepsilon},$$

and therefore

$$g_{n,1}(z_\varepsilon) = \theta_{n,1}\sqrt{d_n(z_\varepsilon)} \le (\theta_\star - \delta)\sqrt{d_n(z_\varepsilon)} \le \sqrt{1 - 3\eta_{\delta,\varepsilon}} \le 1 - \frac{3}{2}\eta_{\delta,\varepsilon}.$$

By Proposition 2.8, the preceding event has probability $1 - o(1)$.

By Lemma 5.2 applied on J_ε, there exist a deterministic constant $C_{\delta,\varepsilon} > 0$ and an integer $n_0(\delta, \varepsilon)$ such that for all $n \ge n_0(\delta, \varepsilon)$, on the event

$$\|X_n\|_{\text{op}} \le b + \frac{\varepsilon}{2}$$

and the compression event on J_ε,

$$\sup_{z \in J_\varepsilon} \left\| H_n(z) - H_n^{(0)}(z) \right\|_{\text{op}} \le \widetilde{\epsilon}_{n,\varepsilon}, \qquad \widetilde{\epsilon}_{n,\varepsilon} := C_{\delta,\varepsilon}\beta_n(J_\varepsilon).$$

In particular, since $\widetilde{\epsilon}_{n,\varepsilon} \to 0$,

$$\lambda_1(H_n(z_\varepsilon)) \le \lambda_1(H_n^{(0)}(z_\varepsilon)) + \widetilde{\epsilon}_{n,\varepsilon} = g_{n,1}(z_\varepsilon) + \widetilde{\epsilon}_{n,\varepsilon} < 1$$

with probability $1 - o(1)$. Lemma 3.7 then implies that, with probability $1 - o(1)$, there are no singular values of W_n above z_ε. Hence

$$\mathbb{P}(\|W_n\|_{\text{op}} \le b + \varepsilon) \to 1.$$

□

6 Proof of Theorem 4.6

Proof of Theorem 4.6. Fix $\varepsilon > 0$ and let $f \in C_c((\gamma_\varepsilon, \infty))$. Extend f by zero to a function $\bar{f} \in C_c((0, \infty))$. Because $\operatorname{supp}(f) \subset\subset (\gamma_\varepsilon, \infty)$, there exists $\tau > 0$ such that

$$\operatorname{supp}(\bar{f}) \subset [\gamma_\varepsilon + 2\tau, \infty).$$

Pick $\delta > 0$ so small that

$$t_\delta^2 < \gamma_\varepsilon.$$

This is possible because $\rho(\theta) \downarrow b$ as $\theta \downarrow \theta_\star$.

Set $k_n := k_{n,\delta}$, and define

$$\eta_n := \begin{cases} \max_{1\le i\le k_n} |\lambda_i(W_n^* W_n) - \Lambda(\theta_{n,i})|, & k_n \ge 1, \\ 0, & k_n = 0. \end{cases}$$

By Theorem 4.3,

$$\eta_n \to 0 \qquad \text{in probability,}$$

and also

$$\lambda_{k_n+1}(W_n^* W_n) = \sigma_{k_n+1}(W_n)^2 \le t_\delta^2 + o_{\mathbb{P}}(1) < \gamma_\varepsilon \qquad \text{in probability.}$$

Hence the event

$$E_n := \left\{\eta_n \le \frac{\tau}{2}\right\} \cap \left\{\lambda_{k_n+1}(W_n^* W_n) < \gamma_\varepsilon\right\}$$

satisfies $\mathbb{P}(E_n) \to 1$.

Work on the event E_n. Because $\lambda_{k_n+1}(W_n^* W_n) < \gamma_\varepsilon$, every actual eigenvalue with index $j > k_n$ gives no contribution to $\int f\, d\mu_{n,\varepsilon}$. Because $\theta_{n,i} < \theta_\star + \delta$ for every spike index $i > k_n$, every such spike satisfies

$$\Lambda(\theta_{n,i}) \le t_\delta^2 < \gamma_\varepsilon,$$

and hence gives no contribution to $\sum_i \bar{f}(\Lambda(\theta_{n,i}))$. Thus only the common index set $1 \le i \le k_n$ remains.

Now let $1 \le i \le k_n$. If $\Lambda(\theta_{n,i}) \le \gamma_\varepsilon + \tau$, then $\bar{f}(\Lambda(\theta_{n,i})) = 0$, and

$$\lambda_i(W_n^* W_n) \le \Lambda(\theta_{n,i}) + \eta_n \le \gamma_\varepsilon + \frac{3\tau}{2} < \gamma_\varepsilon + 2\tau,$$

so also $\bar{f}(\lambda_i(W_n^* W_n)) = 0$. Therefore only indices with

$$\Lambda(\theta_{n,i}) > \gamma_\varepsilon + \tau$$

can contribute. For those indices,

$$|\bar{f}(\lambda_i(W_n^* W_n)) - \bar{f}(\Lambda(\theta_{n,i}))| \le \omega_{\bar{f}}(\eta_n),$$

where $\omega_{\bar{f}}$ is a modulus of continuity of $\bar{f}$. Summing and dividing by r_n yields, on E_n,

$$\left| \int f\, d\mu_{n,\varepsilon} - \frac{1}{r_n} \sum_{i=1}^{r_n} \bar{f}(\Lambda(\theta_{n,i})) \right| \le \omega_{\bar{f}}(\eta_n).$$

Since $\eta_n \to 0$ in probability and $\omega_{\bar{f}}(u) \to 0$ as $u \downarrow 0$, it follows that

$$\omega_{\bar{f}}(\eta_n) \to 0 \qquad \text{in probability.}$$

Hence

$$\int f\, d\mu_{n,\varepsilon} = \int \bar{f}(\Lambda(\theta))\, H_n(d\theta) + o_{\mathbb{P}}(1).$$

Finally, $\bar{f} \circ \Lambda$ is bounded and continuous on $(0, \infty)$, so the weak convergence in probability $H_n \Rightarrow H$ implies

$$\int \bar{f}(\Lambda(\theta))\, H_n(d\theta) \to \int \bar{f}(\Lambda(\theta))\, H(d\theta) \qquad \text{in probability.}$$

Because $\bar{f} = f$ on $(\gamma_\varepsilon, \infty)$, the right-hand side is exactly

$$\int f \, d\big((\Lambda_\# H)|_{(\gamma_\varepsilon,\infty)}\big) = \int f \, d\mu_\varepsilon.$$

This proves the theorem. $\square$

Proof of Corollary 4.8. Set

$$\nu_n := \Lambda_\# H_n, \qquad \nu := \Lambda_\# H, \qquad p_\varepsilon := \nu((\gamma_\varepsilon, \infty)).$$

By Lemma 2.10,

$$\nu_n \Rightarrow \nu \qquad \text{weakly in probability.}$$

Fix $\eta > 0$. Choose $\tau > 0$ such that

$$\nu([\gamma_\varepsilon - \tau, \gamma_\varepsilon + \tau]) < \eta$$

and, in addition,

$$\nu(\{\gamma_\varepsilon - \tau\}) = \nu(\{\gamma_\varepsilon + \tau\}) = 0.$$

This is possible because $\nu(\{\gamma_\varepsilon\}) = 0$ and the set of atoms of a finite measure is countable.

Let

$$N_{n,\varepsilon} := \frac{1}{r_n} \#\{j : \lambda_j(W_n^* W_n) > \gamma_\varepsilon\}.$$

Choose $\delta > 0$ so small that $t_\delta^2 < \gamma_\varepsilon$. By Theorem 4.3, with probability $1 - o(1)$ we are on an event on which

$$\max_{1 \le i \le k_{n,\delta}} |\lambda_i(W_n^* W_n) - \Lambda(\theta_{n,i})| \le \tau$$

and, simultaneously,

$$\lambda_{k_{n,\delta}+1}(W_n^* W_n) < \gamma_\varepsilon.$$

On this event, every spike with $\Lambda(\theta_{n,i}) > \gamma_\varepsilon + \tau$ contributes an actual outlier above γ_ε, while every actual outlier above γ_ε must come from a spike with $\Lambda(\theta_{n,i}) > \gamma_\varepsilon - \tau$. Therefore

$$\nu_n((\gamma_\varepsilon + \tau, \infty)) \le N_{n,\varepsilon} \le \nu_n((\gamma_\varepsilon - \tau, \infty)).$$

Because $\gamma_\varepsilon \pm \tau$ are continuity points of ν, Lemma 2.9 gives

$$\nu_n((\gamma_\varepsilon + \tau, \infty)) \to \nu((\gamma_\varepsilon + \tau, \infty)), \qquad \nu_n((\gamma_\varepsilon - \tau, \infty)) \to \nu((\gamma_\varepsilon - \tau, \infty))$$

in probability. Since

$$\nu((\gamma_\varepsilon - \tau, \infty)) - \nu((\gamma_\varepsilon + \tau, \infty)) \le \nu([\gamma_\varepsilon - \tau, \gamma_\varepsilon + \tau]) < \eta,$$

the squeeze above shows that

$$N_{n,\varepsilon} \to p_\varepsilon \qquad \text{in probability.}$$

Theorem 4.6 gives vague convergence in probability of $\mu_{n,\varepsilon}$ to μ_ε. Since the total masses converge in probability to $\mu_\varepsilon((\gamma_\varepsilon, \infty)) = p_\varepsilon$, Lemma 2.12 applied on the open interval $(\gamma_\varepsilon, \infty)$ yields the weak convergence of $\mu_{n,\varepsilon}$ to μ_ε in probability.

If $p_\varepsilon > 0$, then $\mu_{n,\varepsilon}((\gamma_\varepsilon, \infty)) > 0$ with probability $1 - o(1)$, so the normalized-law statement follows from weak convergence of the numerator and convergence of the normalizing mass. $\square$

7 Haar-singular-vector ensembles

7.1 The ensemble

Recall that

$$\mathrm{St}(N, r) := \{Q \in \mathbb{F}^{N\times r} : Q^*Q = \mathrm{I}_r\}.$$

Assumption 7.1 (Haar-singular-vector noise)**.** Assume that the joint law of (X_n, P_n) admits a representation, still denoted by (X_n, P_n) for simplicity, on some probability space on which

$$X_n = L_n\Sigma_n R_n^*, \qquad \Sigma_n = \mathrm{diag}(s_{n,1}, \dots, s_{n,n}), \qquad s_{n,1} \geq \cdots \geq s_{n,n} \geq 0,$$

with $L_n \in \mathsf{G}(n)$ and $R_n \in \mathrm{St}(m, n)$. Assume further that:

1. P_n is independent of (L_n, Σ_n, R_n);
2. conditional on Σ_n, the random matrices L_n and R_n are independent;
3. conditional on Σ_n, L_n is Haar on $\mathsf{G}(n)$ and R_n is Haar on $\mathrm{St}(m, n)$;
4. the empirical measure of Σ_n^2 converges weakly in probability to a compactly supported law μ with right edge b^2;
5. $\max_k s_{n,k} \to b$ in probability.

Here

$$\mathsf{G}(k) = \begin{cases} \mathrm{U}(k), & \mathbb{F} = \mathbb{C}, \\ \mathrm{O}(k), & \mathbb{F} = \mathbb{R}. \end{cases}$$

Remark 7.2. The law-level formulation is deliberate. For Gaussian matrices, a measurable singular-value decomposition carries a phase or sign coupling between the left and right singular vectors. The results below depend only on the joint law of (X_n, P_n), so an equal-in-law Haar/Stiefel realization is the natural framework.

Remark 7.3 (Examples). Assumption 7.1 includes:

- the rectangular Gaussian model;
- matrices of the form $X_n = L_nT_nR_n^*$ with deterministic diagonal T_n and independent Haar/Stiefel L_n, R_n;
- more generally, any ensemble whose conditional law given its singular values has independent Haar/Stiefel singular vectors.

7.2 Verification of the abstract assumptions

Assumption 7.1(4)–(5) immediately imply Assumption 2.2(1). For the outlier theorems we additionally retain the standing edge hypothesis

$$d(b^+) \in (0, \infty)$$

for the limiting law μ in Assumption 7.1(4). The nontrivial point is Assumption 2.13. We prove it in Appendix A. The statement is recorded here.

Theorem 7.4 (Compression bounds for Haar-singular-vector ensembles)**.** *Assume Assumptions 2.1 and 7.1. Let $I \subset\subset (b, \infty)$ be compact. Then there exists a constant $C_I > 0$ such that*

$$\beta_n(I) := C_I \left(\sqrt{\frac{r_n}{n}} + \sqrt{\frac{\log n}{n}} \right)$$

satisfies Assumption 2.13.

Proof. See Appendix A. The proof uses concentration of quadratic and bilinear forms of Haar/Stiefel vectors, an ε-net upgrade to operator norm, and a Lipschitz-in-z net argument. □

Corollary 7.5 (Main theorem for Haar-singular-vector ensembles)**.** *Under Assumptions 2.1 and 7.1, and assuming in addition that the limiting law μ from Assumption 7.1(4) satisfies $d(b^+) \in (0, \infty)$, Theorems 4.1 and 4.3 hold. In particular,*

$$\max_{1 \le i \le k_{n,\delta}} |\sigma_i(W_n) - \rho(\theta_{n,i})| \le C_\delta \left(\sqrt{\frac{r_n}{n}} + \sqrt{\frac{\log n}{n}} + \zeta_{n,\delta} \right)$$

with probability $1 - o(1)$. If, in addition, $H_n \Rightarrow H$ weakly in probability, then Theorem 4.6 and Corollary 4.7 hold. If moreover

$$H\big(\{\theta : \Lambda(\theta) = (b + \varepsilon)^2\}\big) = 0,$$

then Corollary 4.8 also holds.

8 The Gaussian Marchenko–Pastur example

Assumption 8.1 (Rectangular Gaussian noise)**.** Let $X_n \in \mathbb{F}^{n \times m}$ have i.i.d. entries with law $\mathcal{CN}(0, 1/m)$ in the complex case or $\mathcal{N}(0, 1/m)$ in the real case. Assume $n/m \to c \in (0, 1]$.

Lemma 8.2 (Orbit measure for fixed singular values)**.** *Fix $n \le m$ and a diagonal matrix*

$$\Sigma = \operatorname{diag}(s_1, \ldots, s_n), \qquad s_1 \ge \cdots \ge s_n \ge 0,$$

and let

$$E_0 := \begin{pmatrix} \mathrm{I}_n \\ 0 \end{pmatrix} \in \mathbb{F}^{m \times n}.$$

The map

$$\Psi_\Sigma : \mathsf{G}(n) \times \mathsf{G}(m) \to \mathbb{F}^{n \times m}, \qquad \Psi_\Sigma(L, R) = L \Sigma E_0^* R^*,$$

pushes forward product Haar measure to the unique $\mathsf{G}(n) \times \mathsf{G}(m)$-invariant probability measure on the orbit

$$\mathcal{O}_\Sigma := \{L \Sigma E_0^* R^* : L \in \mathsf{G}(n),\ R \in \mathsf{G}(m)\}.$$

Consequently, if X is bi-unitarily invariant in law, then there is a regular conditional distribution of X given its ordered singular values such that, for almost every value Σ of those singular values, the conditional law is the pushforward of product Haar measure under Ψ_Σ.

Proof. The orbit $\mathcal{O}_\Sigma$ is a compact homogeneous space of $\mathsf{G}(n) \times \mathsf{G}(m)$, namely

$$\mathcal{O}_\Sigma \cong (\mathsf{G}(n) \times \mathsf{G}(m))/\mathrm{Stab}(\Sigma),$$

where $\mathrm{Stab}(\Sigma)$ is the stabilizer of ΣE_0^* under the left-right action. The pushforward of product Haar measure under Ψ_Σ is invariant under this action. Uniqueness of invariant probability on a compact homogeneous space then gives the first claim. If X is bi-unitarily invariant, disintegrate its law with respect to the ordered singular-value map. For almost every singular-value vector Σ, the corresponding regular conditional law is invariant under the left-right action on $\mathcal{O}_\Sigma$. By uniqueness of invariant probability on that compact homogeneous space, it equals the pushforward measure just described. □

Proposition 8.3 (Gaussian admits a Haar-singular-vector representation)**.** *Under Assumptions 2.1 and 8.1, the pair (X_n, P_n) admits an equal-in-law realization satisfying Assumption 7.1. Moreover, the limiting law of the squared singular values, equivalently of the eigenvalues of $X_n X_n^*$, is the Marchenko–Pastur law* MP_c *on*

$$[a_-, a_+] = [(1-\sqrt{c})^2, (1+\sqrt{c})^2],$$

and

$$\|X_n\|_{\mathrm{op}} \to 1+\sqrt{c} \qquad \textit{in probability.}$$

Proof. The Marchenko–Pastur law and the convergence of the top singular value to $1+\sqrt{c}$ are classical [17]. For the singular-vector structure, the Gaussian law is bi-invariant under left multiplication by $\mathsf{G}(n)$ and right multiplication by $\mathsf{G}(m)$. Let Σ_n be the diagonal matrix of singular values of X_n in decreasing order. By Lemma 8.2, conditional on Σ_n, the law of X_n is the same as that of $L_n \Sigma_n R_n^*$ with L_n Haar on $\mathsf{G}(n)$ and R_n Haar on $\mathrm{St}(m,n)$, conditionally independent given Σ_n. Hence $L_n \Sigma_n R_n^*$ has the same unconditional law as X_n. Since P_n is independent of X_n by Assumption 2.1, after replacing (X_n, P_n) by the equal-in-law copy $(L_n \Sigma_n R_n^*, P_n)$, Assumption 7.1 holds. □

Proposition 8.4 (Explicit d-transform in the MP case)**.** *Assume Assumption 8.1. Let*

$$a_\pm := (1 \pm \sqrt{c})^2, \qquad b := 1+\sqrt{c}.$$

Then for $z > b$,

$$\varphi(z) = \frac{z^2 + c - 1 - \Delta(z)}{2cz}, \qquad \widetilde{\varphi}(z) = \frac{z^2 - c + 1 - \Delta(z)}{2z},$$

where

$$\Delta(z) := \sqrt{(z^2 - a_-)(z^2 - a_+)}$$

with the positive branch on (b, ∞). Consequently,

$$d(z) = \frac{2}{z^2 - (1+c) + \Delta(z)}.$$

In particular,

$$d(b^+) = \frac{1}{\sqrt{c}}, \qquad \theta_\star = c^{1/4}.$$

Proof. Let m_{MP_c} be the Stieltjes transform of MP_c in the convention

$$m(w) = \int (x-w)^{-1}\,\mu(dx).$$

The classical formula is

$$m_{\mathrm{MP}_c}(w) = \frac{1-c-w+\sqrt{(w-a_-)(w-a_+)}}{2cw} \qquad (w > a_+).$$

Since

$$\varphi(z) = \int \frac{z}{z^2-x}\,\mu(dx) = -z\,m_{\mathrm{MP}_c}(z^2),$$

the displayed formula for φ follows. The identity for $\widetilde{\varphi}$ is then

$$\widetilde{\varphi}(z) = c\,\varphi(z) + (1-c)\frac{1}{z}.$$

Multiplying the two formulas and simplifying gives the expression for $d(z)$. Finally,

$$d(b^+) = \frac{2}{a_+ - (1+c)} = \frac{1}{\sqrt{c}}.$$

$\square$

Proposition 8.5 (Explicit outlier map in the MP case)**.** *Under Assumption 8.1,*

$$\rho(\theta) = \begin{cases} \sqrt{\dfrac{(1+\theta^2)(c+\theta^2)}{\theta^2}}, & \theta > c^{1/4}, \\ 1+\sqrt{c}, & \theta \le c^{1/4}. \end{cases}$$

Equivalently,

$$\Lambda(\theta) = \begin{cases} \theta^2 + (1+c) + \dfrac{c}{\theta^2}, & \theta > c^{1/4}, \\ (1+\sqrt{c})^2, & \theta \le c^{1/4}. \end{cases}$$

Moreover,

$$d(\rho(\theta)) = \frac{1}{\theta^2} \qquad (\theta > c^{1/4}).$$

Proof. Let $\theta > c^{1/4}$ and set

$$\lambda := \theta^2 + (1+c) + \frac{c}{\theta^2}.$$

Then

$$\lambda - (1+c) = \theta^2 + \frac{c}{\theta^2},$$

and

$$(\lambda - a_-)(\lambda - a_+) = \left(\theta^2 + \frac{c}{\theta^2}\right)^2 - 4c = \left(\theta^2 - \frac{c}{\theta^2}\right)^2.$$

Since $\theta > c^{1/4}$, the positive square root is

$$\sqrt{(\lambda - a_-)(\lambda - a_+)} = \theta^2 - \frac{c}{\theta^2}.$$

Therefore Proposition 8.4 yields

$$d(\sqrt{\lambda}) = \frac{2}{\lambda - (1+c) + \sqrt{(\lambda - a_-)(\lambda - a_+)}} = \frac{2}{2\theta^2} = \frac{1}{\theta^2}.$$

Thus $\rho(\theta) = \sqrt{\lambda}$. The piecewise formula for Λ is then immediate from the definition $\Lambda = \rho^2$. $\square$

Corollary 8.6 (Gaussian mesoscopic spikes). *Under Assumptions 2.1 and 8.1, Theorems 4.1 and 4.3 hold with*

$$\theta_\star = c^{1/4}, \qquad \rho(\theta) = \sqrt{\frac{(1+\theta^2)(c+\theta^2)}{\theta^2}}, \qquad (\theta > c^{1/4}),$$

and

$$\beta_n(I) \lesssim \sqrt{\frac{r_n}{n}} + \sqrt{\frac{\log n}{n}}.$$

Hence, for every fixed $\delta > 0$,

$$\max_{1\le i\le k_{n,\delta}} |\sigma_i(W_n) - \rho(\theta_{n,i})| \le C_\delta \left(\sqrt{\frac{r_n}{n}} + \sqrt{\frac{\log n}{n}} + \zeta_{n,\delta} \right)$$

with probability $1 - o(1)$. If, in addition, $H_n \Rightarrow H$ weakly in probability, then Theorem 4.6 and Corollary 4.7 hold. If moreover

$$H\big(\{\theta : \Lambda(\theta) = (1+\sqrt{c}+\varepsilon)^2\}\big) = 0,$$

then Corollary 4.8 also holds.

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A Sphere and Stiefel concentration

This appendix proves Theorem 7.4.

A.1 Basic concentration lemmas

Lemma A.1 (Chi-square concentration)**.** *Let $g \in \mathbb{F}^N$ have i.i.d. standard real or complex Gaussian entries and set $S = \|g\|_2^2$. Then there exist absolute constants $c, C > 0$ such that for $t \in (0,1)$,*

$$\mathbb{P}(|S - N| \geq tN) \leq 2e^{-ct^2 N}.$$

Lemma A.2 (Bernstein for weighted sub-exponential sums)**.** *Let $\xi_1, \ldots, \xi_N$ be independent centered real random variables with $\|\xi_k\|_{\psi_1} \leq K$. For deterministic $a \in \mathbb{R}^N$,*

$$\mathbb{P}\left(\left|\sum_{k=1}^{N} a_k \xi_k\right| \geq t\right) \leq 2\exp\left(-c\min\left\{\frac{t^2}{K^2\|a\|_2^2}, \frac{t}{K\|a\|_\infty}\right\}\right).$$

Remark A.3. The estimates in Lemmas A.1 and A.2 are standard; see, for example, [18, Chapters 2 and 6].

Lemma A.4 (Quadratic form on the sphere)**.** *Let w be uniform on the unit sphere in $\mathbb{F}^N$, and let $D = \operatorname{diag}(d_1, \dots, d_N)$ be self-adjoint with $\|D\|_{\mathrm{op}} \le M$. Set $\bar{d} = \frac{1}{N} \operatorname{tr} D$. Then for every $t > 0$,*

$$\mathbb{P}\big(|w^* D w - \bar{d}| \ge t\big) \le 2 \exp\left(-cN \frac{t^2}{M^2}\right).$$

Proof. Write $w = g / \|g\|_2$ with g standard Gaussian and set $S = \|g\|_2^2$. Also write $\widetilde{d}_k := d_k - \bar{d}$. Since D is self-adjoint diagonal, the coefficients $\widetilde{d}_k$ are real and satisfy

$$|\widetilde{d}_k| \le 2M, \qquad \sum_{k=1}^{N} \widetilde{d}_k^2 \le 4M^2 N.$$

Because $\sum_k \widetilde{d}_k = 0$,

$$w^* D w - \bar{d} = \frac{\sum_k d_k |g_k|^2}{S} - \bar{d} = \frac{\sum_k \widetilde{d}_k |g_k|^2}{S} = \frac{\sum_k \widetilde{d}_k (|g_k|^2 - 1)}{S}.$$

On the event $\{S \in [N/2, 2N]\}$ we therefore have

$$|w^* D w - \bar{d}| \le \frac{2}{N} \left| \sum_k \widetilde{d}_k (|g_k|^2 - 1) \right|.$$

Now $|g_k|^2 - 1$ is centered and sub-exponential with an absolute ψ_1 norm, so Lemma A.2 gives

$$\mathbb{P}\left(\left| \sum_k \widetilde{d}_k (|g_k|^2 - 1) \right| \ge \frac{tN}{2} \right) \le 2 \exp\left(-cN \min\left\{ \frac{t^2}{M^2}, \frac{t}{M} \right\} \right).$$

For $0 < t \le 2M$, the minimum on the right is bounded below by $\frac{1}{2} t^2 / M^2$. For $t > 2M$, the probability on the left is zero because $|w^* D w - \bar{d}| \le 2M$ deterministically. Combining this with Lemma A.1 for the event $\{S \notin [N/2, 2N]\}$ yields the stated bound after adjusting constants. □

Lemma A.5 (Bilinear form on two independent spheres)**.** *Let w_1, w_2 be independent and uniform on the unit spheres in $\mathbb{F}^{N_1}$ and $\mathbb{F}^{N_2}$. If $D \in \mathbb{F}^{N_1 \times N_2}$ satisfies $\|D\|_{\mathrm{op}} \le M$, then for every $t > 0$,*

$$\mathbb{P}\big(|w_1^* D w_2| \ge t\big) \le 2 \exp\left(-c \, \min(N_1, N_2) \frac{t^2}{M^2} \right).$$

Proof. If $t > M$, the probability is zero because $|w_1^* D w_2| \le M$ deterministically, so it is enough to treat $0 < t \le M$. By symmetry, it is enough to treat the case $N_2 = \min(N_1, N_2)$. Condition on w_1 and set $a := D^* w_1$. Then $\|a\|_2 \le M$. Write $w_2 = g / \|g\|_2$ with g standard Gaussian in $\mathbb{F}^{N_2}$. On the event $\{\|g\|_2^2 \ge N_2/2\}$,

$$|w_1^* D w_2| = \frac{|a^* g|}{\|g\|_2} \le \sqrt{\frac{2}{N_2}} \, |a^* g|.$$

Conditional on w_1, the scalar $a^* g$ is real or complex Gaussian with variance at most $\|a\|_2^2 \le M^2$, so

$$\mathbb{P}\left(|a^* g| \ge t \sqrt{\frac{N_2}{2}} \;\middle|\; w_1 \right) \le 2 \exp\left(-cN_2 \frac{t^2}{M^2} \right).$$

Lemma A.1 gives the same order of bound for the complementary event $\{\|g\|_2^2 < N_2/2\}$. Averaging over w_1 proves the claim. □

Lemma A.6 (ε-net bounds). *Let $\mathcal{N}_\varepsilon$ be an ε-net of the unit sphere in $\mathbb{F}^r$ with $\varepsilon \in (0, 1/4)$. Then*

$$|\mathcal{N}_\varepsilon| \le (1 + 2/\varepsilon)^{2r}.$$

If A is Hermitian $r \times r$,

$$\|A\|_{\mathrm{op}} \le \frac{1}{1-2\varepsilon} \max_{u \in \mathcal{N}_\varepsilon} |u^* A u|,$$

and for general $B \in \mathbb{F}^{r\times r}$,

$$\|B\|_{\mathrm{op}} \le \frac{1}{1-2\varepsilon} \max_{u,v \in \mathcal{N}_\varepsilon} |u^* B v|.$$

Proof. The covering-number bound is standard. For the Hermitian estimate, let x be a unit vector with $|x^* A x| = \|A\|_{\mathrm{op}}$, and choose $u \in \mathcal{N}_\varepsilon$ with $\|x - u\|_2 \le \varepsilon$. Then

$$|x^* A x - u^* A u| \le |(x-u)^* A x| + |u^* A (x-u)| \le 2\varepsilon \|A\|_{\mathrm{op}},$$

which gives

$$\|A\|_{\mathrm{op}} \le \max_{u \in \mathcal{N}_\varepsilon} |u^* A u| + 2\varepsilon \|A\|_{\mathrm{op}}.$$

Rearranging proves the claim. For general B, choose unit vectors x, y with $|x^* B y| = \|B\|_{\mathrm{op}}$, approximate them by $u, v \in \mathcal{N}_\varepsilon$, and use

$$|x^* B y - u^* B v| \le |(x-u)^* B y| + |u^* B (y - v)| \le 2\varepsilon \|B\|_{\mathrm{op}}.$$

This yields the second bound. □

A.2 A representation lemma for Haar Stiefel matrices

Lemma A.7 (Haar-unitary realization of a Haar Stiefel matrix). *Let R be Haar on $\mathrm{St}(m, n)$ and let*

$$E_0 := \begin{pmatrix} \mathrm{I}_n \\ 0 \end{pmatrix} \in \mathbb{F}^{m\times n}.$$

Then there exists a Haar matrix $\widetilde{R} \in \mathsf{G}(m)$ such that

$$R \overset{d}{=} \widetilde{R} E_0.$$

If L is independent Haar on $\mathsf{G}(n)$, then $(L, \widetilde{R})$ may be chosen independent.

Proof. If $\widetilde{R}$ is Haar on $\mathsf{G}(m)$, then $\widetilde{R}E_0$ is Haar on $\mathrm{St}(m, n)$. This realizes the law of R. Independence is obtained by sampling L and $\widetilde{R}$ independently. □

Lemma A.8 (Haar transport of deterministic frames). *Let L be Haar on $\mathsf{G}(N)$ and let $U \in \mathrm{St}(N, r)$ be deterministic. Then $L^* U$ is uniform on $\mathrm{St}(N, r)$. If L_1 and L_2 are independent Haar matrices and U_1, U_2 are deterministic frames, then $L_1^* U_1$ and $L_2^* U_2$ are independent uniform Stiefel matrices.*

Proof. Fix $W \in \mathsf{G}(N)$. Since LW^* is Haar whenever L is Haar, we have

$$W(L^* U) = (LW^*)^* U \overset{d}{=} L^* U.$$

Thus the law of $L^* U$ is left-invariant on $\mathrm{St}(N, r)$, which characterizes the Haar Stiefel law. The independence statement is immediate from the independence of L_1 and L_2. □

Lemma A.9 (Uniformity of compressed Haar directions)**.** *Let Q be uniform on* $\mathrm{St}(N, r)$ *and let $u \in \mathbb{F}^r$ be deterministic with $\|u\| = 1$. Then Qu is uniform on the unit sphere in $\mathbb{F}^N$. If Q_1, Q_2 are independent uniform Stiefel matrices, then $Q_1 u$ and $Q_2 v$ are independent uniform unit vectors.*

Proof. For any fixed $W \in \mathsf{G}(N)$, left invariance of the Haar Stiefel law gives

$$W(Qu) \stackrel{d}{=} (WQ)u \stackrel{d}{=} Qu.$$

Hence the law of Qu is rotation-invariant on the unit sphere, and therefore uniform. The second statement follows from the independence of Q_1 and Q_2. □

A.3 Fixed-z compression bounds

Lemma A.10 (Diagonal compression)**.** *Let $Q \in \mathrm{St}(N, r)$ be uniform and let $D = \mathrm{diag}(d_1, \dots, d_N)$ be deterministic and self-adjoint with $\|D\|_{\mathrm{op}} \le M$. Set $\bar{d} = \frac{1}{N} \operatorname{tr} D$. Then there exist absolute constants $c, C > 0$ such that for every $u \ge 0$,*

$$\mathbb{P}\left(\left\| Q^* D Q - \bar{d}\, \mathrm{I}_r \right\|_{\mathrm{op}} \ge CM\sqrt{\frac{r}{N}} + u \right) \le 2 \exp\left(-cN \frac{u^2}{M^2} \right).$$

Proof. Set

$$A := Q^* D Q - \bar{d}\, \mathrm{I}_r.$$

Fix $\varepsilon = 1/4$ and let $\mathcal{N}$ be a $1/4$-net of the unit sphere in $\mathbb{F}^r$. By Lemma A.6,

$$\|A\|_{\mathrm{op}} \le 2 \max_{x \in \mathcal{N}} |x^* A x|.$$

For fixed $x \in \mathcal{N}$,

$$x^* A x = (Qx)^* D (Qx) - \bar{d}.$$

Hence Lemma A.4 yields

$$\mathbb{P}\big(|x^* A x| \ge s\big) \le 2 \exp\left(-cN \frac{s^2}{M^2} \right) \qquad (s \ge 0).$$

A union bound therefore gives

$$\mathbb{P}\big(\|A\|_{\mathrm{op}} \ge 2s\big) \le 2|\mathcal{N}| \exp\left(-cN \frac{s^2}{M^2} \right).$$

Since $|\mathcal{N}| \le 9^{2r}$, choosing

$$s := C_0 M \sqrt{\frac{r}{N}} + \frac{u}{2}$$

with C_0 sufficiently large yields

$$2r \log 9 - cN \frac{s^2}{M^2} \le -c_1 N \frac{u^2}{M^2}$$

for some absolute $c_1 > 0$. Thus

$$\mathbb{P}\left(\|A\|_{\mathrm{op}} \ge 2C_0 M \sqrt{\frac{r}{N}} + u \right) \le 2 \exp\left(-c_1 N \frac{u^2}{M^2} \right).$$

Renaming constants proves the lemma. □

Lemma A.11 (Mixed compression). *Let $Q_1 \in \mathrm{St}(N_1, r)$ and $Q_2 \in \mathrm{St}(N_2, r)$ be independent uniform Stiefel matrices, and let $D \in \mathbb{F}^{N_1 \times N_2}$ with $\|D\|_{\mathrm{op}} \le M$. Then there exist absolute constants $c, C > 0$ such that for every $u \ge 0$,*

$$\mathbb{P}\left(\|Q_1^* D Q_2\|_{\mathrm{op}} \ge CM\sqrt{\frac{r}{\min(N_1, N_2)}} + u\right) \le 2\exp\left(-c\,\min(N_1, N_2)\frac{u^2}{M^2}\right).$$

Proof. Set

$$B := Q_1^* D Q_2.$$

Fix $\varepsilon = 1/4$ and let $\mathcal{N}$ be a $1/4$-net of the unit sphere in $\mathbb{F}^r$. By Lemma A.6,

$$\|B\|_{\mathrm{op}} \le 2 \max_{x,y \in \mathcal{N}} |x^* B y|.$$

For fixed $x, y \in \mathcal{N}$,

$$x^* B y = (Q_1 x)^* D (Q_2 y).$$

By Lemma A.9, the vectors $Q_1 x$ and $Q_2 y$ are independent uniform unit vectors in $\mathbb{F}^{N_1}$ and $\mathbb{F}^{N_2}$ respectively, so Lemma A.5 gives

$$\mathbb{P}\big(|x^* B y| \ge s\big) \le 2\exp\left(-c\,\min(N_1, N_2)\frac{s^2}{M^2}\right) \qquad (s \ge 0).$$

Applying a union bound over the $|\mathcal{N}|^2 \le 9^{4r}$ pairs (x, y), we obtain

$$\mathbb{P}\big(\|B\|_{\mathrm{op}} \ge 2s\big) \le 2 \cdot 9^{4r} \exp\left(-c\,\min(N_1, N_2)\frac{s^2}{M^2}\right).$$

Now choose

$$s := C_0 M\sqrt{\frac{r}{\min(N_1, N_2)}} + \frac{u}{2}$$

with C_0 sufficiently large to absorb the net cardinality. The same calculation as in Lemma A.10 then gives

$$\mathbb{P}\left(\|B\|_{\mathrm{op}} \ge 2C_0 M\sqrt{\frac{r}{\min(N_1, N_2)}} + u\right) \le 2\exp\left(-c_1\,\min(N_1, N_2)\frac{u^2}{M^2}\right).$$

Renaming constants proves the result. □

A.4 Uniformity in z

Lemma A.12 (Lipschitz lift). *Let $F : I \to \mathbb{F}^{N_1 \times N_2}$ satisfy*

$$\big\|F(z) - F(z')\big\|_{\mathrm{op}} \le L|z - z'| \qquad (z, z' \in I).$$

Then for any Stiefel matrices Q_1, Q_2,

$$\big\|Q_1^* F(z) Q_2 - Q_1^* F(z') Q_2\big\|_{\mathrm{op}} \le L|z - z'|.$$

Proof. Use $\|Q_1\|_{\mathrm{op}} = \|Q_2\|_{\mathrm{op}} = 1$. □

Proof of Theorem 7.4. Fix a compact interval $I = [z_-, z_+]$ with $z_- > b$. Let

$$\eta_I := \frac{z_- - b}{2} > 0, \qquad \mathcal{E}_{n,I} := \Big\{ \max_k s_{n,k} \le b + \eta_I \Big\}.$$

By Assumption 7.1(5),

$$\mathbb{P}(\mathcal{E}_{n,I}) \to 1.$$

Since $b + \eta_I = (b + z_-)/2 < z_-$, the event $\mathcal{E}_{n,I}$ implies $\|X_n\|_{\mathrm{op}} < \inf I$.

Using the representation from Assumption 7.1, and then Lemma A.7, we may without loss of generality work on a probability space on which

$$X_n = L_n \Sigma_n E_0^* \widetilde{R}_n^*,$$

where $\widetilde{R}_n$ is Haar on $\mathsf{G}(m)$ and, conditional on Σ_n, the matrices L_n and $\widetilde{R}_n$ are independent and independent of P_n.

Define the diagonal matrices

$$\begin{aligned} A_n(z) &:= \operatorname{diag}\big(a_{n,1}(z), \dots, a_{n,n}(z)\big), \\ \widetilde{A}_n(z) &:= \operatorname{diag}\big(a_{n,1}(z), \dots, a_{n,n}(z), z^{-1}, \dots, z^{-1}\big), \\ B_n(z) &:= \begin{pmatrix} \operatorname{diag}\big(b_{n,1}(z), \dots, b_{n,n}(z)\big) & 0 \end{pmatrix}, \end{aligned}$$

where

$$a_{n,k}(z) := \frac{z}{z^2 - s_{n,k}^2}, \qquad b_{n,k}(z) := \frac{s_{n,k}}{z^2 - s_{n,k}^2}.$$

Here the zero block in $B_n(z)$ has size $n \times (m - n)$. Let

$$\mathcal{F}_n := \sigma(\Sigma_n, U_n, V_n).$$

Because $(L_n, \Sigma_n, \widetilde{R}_n)$ is independent of P_n, conditional on $\mathcal{F}_n$ the matrices L_n and $\widetilde{R}_n$ have the same law as conditional on Σ_n alone. By Lemma A.8, conditional on $\mathcal{F}_n$ the random matrices

$$Q_{U,n} := L_n^* U_n \in \mathrm{St}(n, r_n), \qquad Q_{V,n} := \widetilde{R}_n^* V_n \in \mathrm{St}(m, r_n)$$

are independent uniform Stiefel matrices. Moreover,

$$\begin{aligned} \mathcal{A}_n(z) &= Q_{U,n}^* A_n(z) Q_{U,n}, \\ \mathcal{D}_n(z) &= Q_{V,n}^* \widetilde{A}_n(z) Q_{V,n}, \\ \mathcal{B}_n(z) &= Q_{U,n}^* B_n(z) Q_{V,n}, \end{aligned}$$

while

$$\varphi_n(z) = \frac{1}{n} \operatorname{tr} A_n(z), \qquad \widetilde{\varphi}_n(z) = \frac{1}{m} \operatorname{tr} \widetilde{A}_n(z).$$

On the event $\mathcal{E}_{n,I}$ and for $z \in I$, the denominators satisfy

$$z^2 - s_{n,k}^2 \ge z_-^2 - (b + \eta_I)^2 =: \kappa_I > 0.$$

Hence there exists a deterministic constant $M_I < \infty$ such that on $\mathcal{E}_{n,I}$,

$$\sup_{z \in I} \max\Big\{ \|A_n(z)\|_{\mathrm{op}}, \Big\|\widetilde{A}_n(z)\Big\|_{\mathrm{op}}, \|B_n(z)\|_{\mathrm{op}} \Big\} \le M_I.$$

Applying Lemmas A.10 and A.11 conditionally on $\mathcal{F}_n$ therefore yields, for each fixed $z \in I$ and every $u \geq 0$,

$$\mathbb{P}\Big(\|\mathcal{A}_n(z) - \varphi_n(z)\mathrm{I}\|_{\mathrm{op}} > CM_I\sqrt{r_n/n} + u \,\Big|\, \mathcal{F}_n\Big)\mathbf{1}_{\mathcal{E}_{n,I}} \leq 2e^{-cnu^2/M_I^2},$$
$$\mathbb{P}\Big(\|\mathcal{D}_n(z) - \widetilde{\varphi}_n(z)\mathrm{I}\|_{\mathrm{op}} > CM_I\sqrt{r_n/n} + u \,\Big|\, \mathcal{F}_n\Big)\mathbf{1}_{\mathcal{E}_{n,I}} \leq 2e^{-cnu^2/M_I^2},$$
$$\mathbb{P}\Big(\|\mathcal{B}_n(z)\|_{\mathrm{op}} > CM_I\sqrt{r_n/n} + u \,\Big|\, \mathcal{F}_n\Big)\mathbf{1}_{\mathcal{E}_{n,I}} \leq 2e^{-cnu^2/M_I^2}.$$

Here we used $\min(n, m) = n$.

Next, on $\mathcal{E}_{n,I}$ the derivatives of the scalar functions are uniformly bounded on I:

$$|a'_{n,k}(z)| = \frac{z^2 + s_{n,k}^2}{(z^2 - s_{n,k}^2)^2} \leq L_I, \qquad |b'_{n,k}(z)| = \frac{2zs_{n,k}}{(z^2 - s_{n,k}^2)^2} \leq L_I, \qquad \left|\frac{d}{dz}z^{-1}\right| \leq L_I,$$

for a deterministic constant L_I. Hence

$$\big\|A_n(z) - A_n(z')\big\|_{\mathrm{op}} + \Big\|\widetilde{A}_n(z) - \widetilde{A}_n(z')\Big\|_{\mathrm{op}} + \big\|B_n(z) - B_n(z')\big\|_{\mathrm{op}} \leq 3L_I|z - z'|$$

on $\mathcal{E}_{n,I}$. By Lemma A.12, the same is true for the compressed matrices, and by averaging the scalar entries the same is true for φ_n and $\widetilde{\varphi}_n$. Consequently, on $\mathcal{E}_{n,I}$ the three matrix-valued maps

$$z \mapsto \mathcal{A}_n(z) - \varphi_n(z)\mathrm{I}, \qquad z \mapsto \mathcal{D}_n(z) - \widetilde{\varphi}_n(z)\mathrm{I}, \qquad z \mapsto \mathcal{B}_n(z)$$

are uniformly Lipschitz on I with deterministic constant C_I^{Lip}, and therefore

$$\Xi_n(z) := \|\mathcal{A}_n(z) - \varphi_n(z)\mathrm{I}\|_{\mathrm{op}} + \|\mathcal{D}_n(z) - \widetilde{\varphi}_n(z)\mathrm{I}\|_{\mathrm{op}} + \|\mathcal{B}_n(z)\|_{\mathrm{op}}$$

is Lipschitz on I with deterministic constant $L_I^{(\Xi)}$.

Choose

$$u_n := K_I\sqrt{\frac{\log n}{n}}$$

with K_I so large that $cK_I^2 > 2$, and let $\mathcal{Z}_n \subset I$ be a deterministic h_n-net with

$$h_n := \frac{u_n}{4L_I^{(\Xi)}}.$$

Then $|\mathcal{Z}_n| \leq C_I/u_n \leq C_I\sqrt{n/\log n}$. For a fixed $z_0 \in \mathcal{Z}_n$, apply the three fixed-z bounds with threshold $u_n/3$. After enlarging the deterministic constant C_I, a union bound over the three terms gives

$$\mathbb{P}\Big(\Xi_n(z_0) > C_I\sqrt{r_n/n} + u_n \,\Big|\, \mathcal{F}_n\Big)\mathbf{1}_{\mathcal{E}_{n,I}} \leq 6e^{-c_I nu_n^2}.$$

A further union bound over $z_0 \in \mathcal{Z}_n$ yields

$$\mathbb{P}\Big(\sup_{z_0 \in \mathcal{Z}_n} \Xi_n(z_0) > C_I\sqrt{r_n/n} + u_n \,\Big|\, \mathcal{F}_n\Big)\mathbf{1}_{\mathcal{E}_{n,I}} \leq \frac{C_I}{u_n}e^{-c_I nu_n^2} = o(1),$$

where the $o(1)$ is deterministic. Using the Lipschitz bound to pass from $\mathcal{Z}_n$ to all of I costs at most

$$L_I^{(\Xi)}h_n \leq \frac{u_n}{4}.$$

Therefore,

$$\mathbb{P}\Big(\sup_{z\in I}\Xi_n(z) > C_I\big(\sqrt{r_n/n}+u_n\big)\;\Big|\;\mathcal{F}_n\Big)\mathbf{1}_{\mathcal{E}_{n,I}} = o(1)$$

with a deterministic $o(1)$ bound. Taking expectations and using $\mathbb{P}(\mathcal{E}_{n,I}) \to 1$ yields

$$\mathbb{P}\Big(\mathcal{E}_{n,I},\ \sup_{z\in I}\Xi_n(z) > C_I\big(\sqrt{r_n/n}+\sqrt{\log n/n}\big)\Big) \to 0.$$

Since

$$\sup_{z\in I}\|\mathcal{A}_n(z)-\varphi_n(z)\mathrm{I}\|_{\mathrm{op}} + \sup_{z\in I}\|\mathcal{D}_n(z)-\widetilde{\varphi}_n(z)\mathrm{I}\|_{\mathrm{op}} + \sup_{z\in I}\|\mathcal{B}_n(z)\|_{\mathrm{op}} \le 3\sup_{z\in I}\Xi_n(z),$$

we conclude, after absorbing the factor 3 into the constant, that

$$\mathbb{P}\left(\begin{array}{c}\|X_n\|_{\mathrm{op}} < \inf I,\\ \sup_{z\in I}\|\mathcal{A}_n(z)-\varphi_n(z)\mathrm{I}\|_{\mathrm{op}} + \sup_{z\in I}\|\mathcal{D}_n(z)-\widetilde{\varphi}_n(z)\mathrm{I}\|_{\mathrm{op}} + \sup_{z\in I}\|\mathcal{B}_n(z)\|_{\mathrm{op}}\\ \le C_I\big(\sqrt{r_n/n}+\sqrt{\log n/n}\big)\end{array}\right) \to 1.$$

This is exactly Assumption 2.13. $\square$

Acknowledgments

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon 2022 research and innovation program (grant agreement No. 101041711), by the Simons Foundation (as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning), by the Israel Science Foundation (grant number 2258/19), by the Israel Science Foundation (ISF Grant 4101/25), and by the U.S. National Science Foundation (NSF Grant OISE-2401227).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.